Certified Correction of Vision–Language Hallucinations: Gains, Preconditions, and a Sequential Failure Mode

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Abstract

Selective prediction equips a vision–language model (VLM) with a distribution-free guarantee on the answers it emits, and pays for that guarantee by abstaining on the rest. A recent impossibility result shows the bill cannot be reduced by better engineering: when the base risk μ exceeds the target α, *any* predictor that may only *emit or abstain the model's own answer* must abstain on at least $(\mu - \alpha)/(1 - \alpha)$ of inputs, whatever its score, bound, or calibration budget. This paper takes that premise seriously and asks what changes when the predictor is permitted to emit a *different* answer. We introduce Certified-Correction Risk Control (CCRC): accept the model's answer where a correctness score is high, substitute an answer supplied by an *independent* evidence channel where that channel's evidence is strong, abstain otherwise, and certify the error rate of everything emitted at level α with confidence $1 - \delta$.

We prove validity for CCRC over a policy family indexed by pre-specified score *values* (the rank-indexed variant we implement is validated empirically rather than proved), characterise its behaviour, and evaluate it across five benchmarks, two backbones and a mitigation decoder, under a single protocol: a three-way disjoint split, fixed-sequence testing, exact binomial bounds, and ≥ 400 paired splits with intervals and two-sided paired tests on every contrast. Five findings organise the paper, and §1 lists the contributions they support. First, the correctness *score* matters more than the procedure: a detection-grounded score improves certified coverage over model-internal uncertainty by 3.1–45.6 points, and we decompose our total margin over a published CLIP-based scorer to show that 64.5 of 66.8 points come from the score and 2.3 from our algorithm — the 2.3 is the number we ask a reader to trust. Second, self-certified repair is bounded rather than impossible: we give a mixture-level budget showing that substituting the complement of the score that flagged an item can occupy only as much mass as the accepted region's slack, and measure that on the settings where correction pays no region cheap enough exists, so certified coverage collapses by 6.6–49.3 points. Third, where CCRC helps, the gain decomposes into a part that is automatic — the 1.0–2.0% of repaired items are themselves emitted — and a genuine *dilution* bonus that fires only on the 12–29% of splits where the calibrated threshold actually moves and is worth +0.2 to +1.8 points. Certified coverage rises by +0.4 to +7.6 points in four settings, with every interval excluding zero, and falls by 10.2 points on the fifth; the loss is an abort phenomenon of the sequential test under a 76-item calibration fold, not absent headroom. Fourth, a polarity-symmetric repair gate — the obvious remedy for our one-directional evidence channel — is valid and admits negative-polarity repairs at 99.9–100% precision, and yet corrects 0.000% of false-positive existence hallucinations in all ten cells, because on that side repair opportunity and repair precision are anti-correlated. Fifth, the natural generative extension does not work: in a paired matched-prefix experiment, repairing a claim mid-caption *increases* downstream hallucination ($+0.207 \pm 0.091$ mentions, $n = 121$, two-sided $p = 0.025$, though under-powered as a magnitude claim), which withdraws support for a fixed-point calibration argument (correction-map monotonicity, not the risk monotonicity in λ that our one-shot guarantee avoids needing) and exposes an unpriced cost that a context-blind sequential procedure must charge for. We also reproduce the closest concurrent method

and find the two mechanisms are complementary rather than competing. Code, extracted per-item signals, and all negative results are released.

1 Introduction

1.1 A small failure, and three unsatisfying options

Image 518 of the POPE benchmark is an office desk. At its lower right, small but unoccluded and plainly visible, is a computer mouse. We ask LLaVA-1.5-7B: *"Is there a mouse in the image?"* The model answers **no**, and it is not especially confident about it — the probability mass on `yes` is 0.207. Our calibrated correctness score for this item is $s = 0.18$, at roughly the 0.2nd percentile of the score distribution. An open-vocabulary detector, asked independently whether it can localise "a photo of a mouse" in the same pixels, returns a confidence of 0.903 with a margin of 0.753 over its own negative-class baseline. The object is there. The detector sees it. The model does not.

Now suppose we must ship this system with a promise: *among the answers we actually emit, at most* 10% *are wrong, with* 90% *confidence, and this must hold for the next batch of inputs rather than the one we tuned on.* Such promises are what conformal risk control was built to deliver, and the machinery is by now standard (Vovk et al., 2005; Bates et al., 2021; Angelopoulos et al., 2021). The standard recipe has exactly one lever: a threshold on a correctness score, calibrated on held-out data, above which we emit the model's answer and below which we abstain.

Applied to item 518, the recipe has three options, and all three are unsatisfying.

It can **emit** the answer. But $s = 0.18$ is low, and if we set the threshold low enough to admit this item we also admit a large population of items that genuinely are uncertain and genuinely are wrong; the risk budget is consumed and the guarantee fails.

It can **abstain**. This is what every conformal method for vision–language models we are aware of does, and it is safe. It is also, here, absurd: we are declining to answer a question to which a component already inside our own system supplies the correct answer with high confidence. We pay a coverage cost for an item we could have got right.

It can **delete the claim** — the claim-filtering move used in the generative setting (Li et al., 2025; Mohri & Hashimoto, 2024). In a yes/no setting deletion is abstention. In a captioning setting it is slightly better than abstention and considerably worse than what a reader wants, since a caption with a hole in it is not a caption that has been corrected.

What none of the three can do is emit **yes**. That option is not missing because it was tried and found wanting. It is missing because the framework does not contain it. The action space of a conformal selective predictor is $\{\text{emit the model's answer}, \text{abstain}\}$, and this paper is about what happens when a third action — *emit a different answer, and certify that too* — is added to it.

1.2 Two literatures that do not meet

The reason the third action is absent is, we think, sociological as much as technical: the two communities with a stake in it work on different halves of the problem.

One literature makes the model wrong less often, and is inventive about it: modified decoding (Huang et al., 2024; Leng et al., 2024; Zhang et al., 2025), mechanistic and positional interventions (Jiang et al., 2025; Wan et al., 2025; Deng & Yang, 2025; Cheng et al., 2026; Zhao et al., 2025b), injected vision-expert guidance (Zhao et al., 2025a), post-hoc verify-and-rewrite (Yin et al., 2024; Zhou et al., 2024; Wu et al., 2025), and retrieval grounding (Lee et al., 2024; Xia et al., 2024). The measured effects are substantial and the methods are cheap. But its output is always of the form *our hallucination rate is lower than theirs on this benchmark*, and a deployer who asks "what fraction of the claims you emit tomorrow will be false?" gets an average over a fixed test set, which is not an answer to the question asked.

The other literature answers that question and charges for it. Conformal prediction and its risk-control descendants turn any heuristic score into a decision rule with a finite-sample, distribution-free guarantee; applied to generation the result is *selective prediction*, in which the error rate among emitted outputs is at most α with probability $1-\delta$ (Quach et al., 2024; Mohri & Hashimoto, 2024; Li et al., 2025; Lee et al., 2025). The price is paid in coverage: every item the rule cannot vouch for is an item nobody is served.

Neither literature is wrong, but notice what falls between them. The first builds systems that can *repair* a wrong answer and never certifies the result; the second certifies rigorously and never repairs. The obvious composition, *certified repair*, is what the verify-and-correct line (Yin et al., 2024; Zhou et al., 2024; Dang et al., 2026) gestures at without achieving. This paper asks what it would take to attach a real guarantee to a system that changes its own answers.

1.3 The price of a guarantee, and why it is not a tuning problem

Before asking that, it is worth being precise about how expensive abstention is, because the size of the number is what makes the third action interesting rather than merely tidy.

Kotte (2026a) give the answer in closed form. If the base risk of the underlying model is μ and the target is $\alpha < \mu$, then *any* conformal selective predictor must abstain on at least

$$\frac{\mu-\alpha}{1-\alpha}$$

of inputs. The bound does not depend on the quality of the score, the tightness of the concentration inequality, or the size of the calibration set. A perfect oracle score obeys it. A larger calibration set does not relax it. It is a property of the action space, not of the estimator.

The consequences on real systems are not subtle. On our POPE-1500 setup with LLaVA-1.5-7B the empirical base risk is $\mu = 18.1\%$; at $\alpha = 0.10$ at least 9.0% of inputs must be refused before any method is chosen. Our own tuned filtering pipeline, which is competitive with published conformal VLM results, certifies 44.0% coverage on POPE-adversarial at $\alpha = 0.10$: three in five questions go unanswered. On HallusionBench, where the base risk is high enough that the floor swallows the entire budget, it certifies 0.0%. Zero. The guarantee holds vacuously, over an empty emitted set, which is exactly the failure mode a practitioner would call useless.

The floor is not what binds today, and we do not pretend otherwise. Those two numbers invite an overattribution we want to head off, because it would flatter this paper. The floor on POPE-adversarial is 8.1 points; the abstention our pipeline actually incurs there is 56.0 points. Across our five settings the floor accounts for only 14–29% of realised abstention (§3.3). The remaining three-quarters is score quality and finite-sample power — the very things the theorem says it cannot speak to. The floor is therefore the right reason to look at the *action space*: it is the one part of the bill no amount of engineering will retire, and it is why the substitution action is worth introducing at all. It is not a claim that removing it recovers the other 40 points. That is exactly why the honest accounting in §5.6 attributes 64.5 points to the correctness score and 2.3 to our own mechanism, and why we report the 2.3.

This is also the point at which the natural engineering instinct — *find a better score* — runs into the theorem. A better score moves you toward the floor faster. It cannot move you through it. Whatever is to be done *to the floor* must be done to the action space.

1.4 Re-reading the premise

So we re-read the premise. Kotte (2026a) state it plainly: the predictor may only emit the model's own answer or abstain. Every conformal method for VLMs we surveyed satisfies it, and satisfies it structurally rather than incidentally. Claim-filtering deletes unsupported claims (Li et al., 2025; Mohri & Hashimoto, 2024). Abstention methods decline (Lee et al., 2025; Wagner, 2026). The closest concurrent work, budgeted conformal evidence acquisition (Xu et al., 2026), acquires *additional visual evidence* for borderline claims and re-scores the original assertion — a genuinely clever move that buys real coverage, and one that explicitly leaves the emitted answer unchanged. In every case the object emitted is the model's output or nothing.

Item 518 says that this is a choice, not a necessity. Our system already contains a component that knows the right answer. The question is whether we can emit that answer *and still have a guarantee.*

What changes. Permitting substitution does not evade the mathematics; it changes the object being certified. If the emitted answer may originate outside the model, the quantity to control is the risk of a *mixture* — accepted model answers together with substituted ones — and the bound of Kotte (2026a) no longer binds, because its premise is violated. The guarantee must be re-derived for the mixture, which we do (Theorem 1), and the substitutions must come from somewhere trustworthy, which turns out to be a hard constraint rather than an implementation detail (§4.4).

Why it can win, in one sentence. High-precision repairs are *low-risk* emissions, so adding them to the emitted set pulls its average risk below α, which slackens the binding constraint and can let the acceptance threshold move down, admitting a larger population of ordinary accepted items. We are careful about how much of the measured gain that mechanism actually explains (§4.7): the 1.0–2.0% of repaired items are emitted whether or not the threshold moves, which is arithmetic rather than leverage, and the threshold is *unchanged* on 71–88% of splits. The dilution term proper is worth +0.2 to +1.8 points. So the mechanism is real, measured, and secondary; the headline is that a small mass of near-certain repairs can be emitted essentially free.

1.5 What went wrong on the way, and what it taught us

It would be convenient to report that the idea worked. It works in four settings out of five, and the interesting content of this paper is largely in the failures, each of which turned out to be forced rather than accidental. We list them here because they are the spine of the argument, and each is a section below.

The cheap version defeats itself. The first design certified repairs using the complement of the score that flagged the original: if s is low, surely $1-s$ vouches for flipping the answer. For binary decisions the risk of the flipped answer on a flagged region equals the model's *accuracy* on that region, so a self-repair region is cheap only where the model is right less than α of the time — a strictly stronger demand than finding a region where it is unconfident. It is admissible whenever the accepted region has slack (Proposition 2), but on the settings where correction is worth doing the region does not exist (22.1% accuracy in the bottom 2% of our score against $\alpha = 0.10$; 62.4% on Qwen2-VL), and attempting it makes the fixed sequence abort outright on 25–99% of splits. So a second, independent channel is required in practice — an empirical regularity with a mechanism, not a theorem.

The obvious parameterisation loses on half our backbones. Given two channels, the natural move is to sweep both gates with the single risk-control parameter. We did, and it lost 4.2 points on Qwen2-VL and 2.9 on VCD-decoded LLaVA relative to plain filtering, because a liberal acceptance threshold drags the repair gate open with it and admits low-precision repairs precisely when the budget is tightest. The fix is to decouple: the repair gate is fixed and strict, and only acceptance is calibrated (§4.5). We report the version that failed because the reason it failed is the reason the final design looks the way it does.

The gains are real, modest, and not uniform. Certified coverage rises by +0.4 to +7.6 points in four settings and falls by 10.2 on the fifth. We chased that reversal under the wrong hypothesis — that the achievable gain is capped by the abstained mass, so a benchmark already certifying most of its inputs has little to gain (§4.8). The bound is true and useless as an explanation: conditional on both arms certifying anything, CCRC *wins* on the losing benchmark (+2.17 points), dilution runs in its favour there, and the entire reported loss is the fixed sequence aborting on 13% of splits under a 76-item calibration fold — the same prefix mechanism that defeats self-repair (§6). Upside is bounded, downside is not, and what binds is the test's behaviour at small emitted counts rather than headroom.

Most of our headline margin is not our algorithm. Compared against a published conformal VLM scorer we gain 66.8 points of certified coverage on one setting. Decomposed, 64.5 of those points come from the correctness score — a boring combination of logit and grounding features — and 2.3 from the certified-correction mechanism itself (§5.6). We would rather a reader trust the 2.3 than be sold the 67.

The obvious remedy for our one-directional gate remedies nothing. Our evidence channel can only argue that an object is *present*, so as evaluated CCRC repairs missed objects and never the canonical false-positive existence hallucination (Remark 1). We built the polarity-symmetric gate that fixes this in principle. It is valid, it admits negative-polarity repairs at 99.9–100% precision, and it corrects 0.000% of false-positive hallucinations in all ten cells, because on the negative side the items reachable at high precision are the ones the model already answers correctly (§5.3). An admitted gap becomes a measured impossibility.

In generation, correction backfires. The motivating case for repair is generative. Extending the guarantee to a sequence requires that repairing a claim not make later claims worse — a monotonicity property that would license a fixed-point argument. A paired matched-prefix test finds the opposite sign: repaired prefixes yield $+0.207 \pm 0.091$ additional hallucinated mentions downstream ($n = 121$, two-sided $p = 0.025$). The experiment is directional but under-powered (§7.1), so the supported conclusion is that our data does not support the fixed-point route, not that the route is closed. We give the coupling bound that survives (Theorem 2).

The closest competitor turns out to be an ally. Concurrent work (Xu et al., 2026) shares our benchmarks, machinery and motivation. Reproducing it under a common base score shows the two mechanisms rescue disjoint populations of claims and compose additively (§5.7), so we narrow our novelty claim to the substitution action and its guarantee.

The paper is arranged so that each design decision is forced by a measurement rather than asserted, and the negative results appear as findings rather than as caveats appended to a positive headline.

1.6 Contributions

1. **A procedure and its guarantee** (§4). CCRC emits accepted, repaired, or abstained outputs and certifies the risk of the emitted mixture: $\Pr[\mathcal{R}(\mathcal{E}) \leq \alpha] \geq 1 - \delta$ (Theorem 1), using fixed-sequence testing over a nested policy family with exact binomial bounds. We also give the minimum certifiable sample size (Lemma 1), a small result with practical teeth: a fixed sequence started below it aborts and returns zero coverage. The value is bound-dependent — 22 for Clopper–Pearson but 44, 71 and 116 for the betting, empirical-Bernstein and Hoeffding bounds at $\alpha = \delta = 0.10$ — so a grid start tuned for one bound silently disables another.

2. **Why repair cannot be self-certified in practice** (§4.4). Certifying a substitution using the complement of the score that flagged the original makes the repair's risk equal to the model's *accuracy* on the flagged region. Proposition 2 bounds the mass such a repair may occupy by the accepted region's slack; we then measure that on the settings where correction is worth doing no sufficiently low-accuracy region exists, and that the fixed-sequence test aborts entirely on 25–99% of splits when self-repair is attempted. The conclusion — use a second, independent channel — is an empirical regularity with a mechanism, not a theorem.

3. **A measured mechanism, and a precondition we withdraw as an explanation** (§4.7, §4.8). High-precision repairs lower the emitted set's average risk, relaxing the binding constraint and permitting a more permissive acceptance threshold (Proposition 3). We then decompose the realised gain and find that most of it is not that mechanism: the repaired 1.0–2.0% is emitted automatically, the calibrated threshold is unchanged on 71–88% of splits, and the dilution term proper is +0.2 to +1.8 points. Proposition 4 bounds the achievable gain by the abstained mass, which is true but, we now show, has *no confirming instance* in our data: the headroom diagnostic built on it is demoted to Conjecture 1 and the one loss it was supposed to explain is explained instead by §6.

4. **A small-calibration failure mode of fixed-sequence testing** (§6). The minimum certifiable emitted-set size (Lemma 1) combines with the prefix structure of the test into a single mechanism

that explains three apparently unrelated results: weak scores certify nothing at all rather than a little less, self-repair collapses, and CCRC's one negative result is an abort phenomenon under a 76-item calibration fold. We give the bound-dependent $k_{\min}$ ladder (22, 44, 71, 116), report abort rates everywhere, and identify the condition (feasibility failing to be a prefix in λ) under which the first hypothesis decides the outcome.

5. **Two measured impossibilities** (§5.3, §7). A polarity-symmetric repair gate is valid at no cost in δ and admits negative-polarity repairs at 99.9–100% precision, yet corrects no false-positive existence hallucination in any setting, because on that side repair opportunity and precision are anti-correlated. And in a paired matched-prefix experiment on free-form captions, repairing a hallucinated claim increases hallucination in the continuation, which withdraws support for the *correction-map* monotonicity a fixed-point argument would need (a different property from risk monotonicity in λ, which Theorem 1 never uses); we give the coupling bound that survives (Theorem 2) and a power analysis bounding what the experiment can and cannot establish.

6. **Honest attribution and positioning** (§5.6). We decompose our margin over a published scorer and report that 64.5 of 66.8 points are the score and 2.3 the procedure. We reproduce the closest concurrent method (Xu et al., 2026) and find the two mechanisms compose rather than compete.

1.7 Roadmap

§2 situates the work between the two literatures above. §3 fixes notation, derives the abstention floor in our own terms, and measures how much of realised abstention it accounts for. §4 gives the procedure (Algorithm 1), its guarantee, the self-certification budget, the decoupling argument, the dilution mechanism and its decomposition, and the precondition. §5 reports experiments across five benchmarks, two backbones and one mitigation decoder, including the symmetric-gate result and the setting where the method fails. §6 consolidates the small-calibration failure mode. §7 reports the sequential falsification, its power, and what survives it. §8 isolates the score combiner (with the validity trap that surrounds it) and the choice of concentration bound. §9 states the limitations we consider load-bearing.

2 Related work

This section is organised around the gap identified in §1: the hallucination-mitigation literature repairs without certifying (§2.1), the distribution-free risk-control literature certifies without repairing (§2.2, §2.3), and the uncertainty-quantification literature (§2.4) supplies scores to both. §2.5 treats the two concurrent papers that bear most directly on our claim and §2.6 the shift induced by acting on one's own output.

2.1 Hallucination in vision–language models

Phenomenon, taxonomy and causes. Hallucination in multimodal models is conventionally decomposed by what is fabricated: *existence* (an absent object is asserted), *attribute* (a present object is described wrongly), and *relation* (spatial or interactional claims that do not hold) (Wang et al., 2023a). Surveys (Bai et al., 2024; Liu et al., 2024b; Ji et al., 2023) converge on a few proximate causes: *language-prior dominance* (the decoder completes "a desk with a keyboard and a. . . " with `mouse` whether or not one is visible), *visual under-attention* (attention mass on image tokens decays with position), *instruction-tuning artefacts* from supervision that describes more than is visible (Liu et al., 2024a), and ordinary *miscalibration* (Guo et al., 2017), which matters for us because it determines how informative the model's own logits are as a correctness score.

The taxonomy is not decoration: it determines which evidence can refute a claim, and therefore which claims a system like ours can repair. Existence claims are checkable by an open-vocabulary detector (Minderer et al., 2023; Liu et al., 2024d); attribute claims require finer perception than a detector supplies; relation claims require structure that neither a detector nor a similarity model represents. §5.4 shows the composition of a benchmark — not its difficulty — deciding whether our evidence channel applies at all.

Benchmarks, and what they measure. POPE (Li et al., 2023) probes existence with balanced yes/no questions under *random*, *popular* and *adversarial* negative samplings; the last is hardest because it pits the language prior against the image, which makes it the most informative split for our purposes. CHAIR (Rohrbach et al., 2018) scores free-form captions by the fraction of mentioned objects absent from MS-COCO ground truth. AMBER (Wang et al., 2023a) provides an LLM-free benchmark with per-image present/hallucinated object lists across generative and discriminative splits. MMHal-Bench, introduced by Sun et al. (2024) alongside Fact-RLHF, uses a strong judge model. MME (Fu et al., 2023) measures perception and cognition across fourteen subtasks, only some grounding-checkable; HallusionBench (Guan et al., 2024) targets illusions, charts and reasoning traps and is hard enough that strong models approach chance — a property that matters a great deal for any method with an abstention floor; GQA (Hudson & Manning, 2019) supplies compositional questions over scene graphs.

We use POPE, AMBER, MME, GQA and HallusionBench. One methodological remark belongs here: single-type AMBER subsets are label-degenerate (the `hallucination` subset has all-`no` gold answers, the `relation` subset all-`yes`), which inflates AUROC to 1.0 for any score correlated with the question type. We report on the mixed subset for this reason, and flag it because the degeneracy is easy to miss.

Mitigation by decoding, intervention and training. The largest and most active family modifies the decoding distribution at inference time with no retraining. OPERA (Huang et al., 2024) identifies a columnar attention pattern over a few *summary* tokens, penalises it in the beam score, and rolls back to re-allocate *token selection*. Visual contrastive decoding (Leng et al., 2024) subtracts logits computed from a diffusion-corrupted image, removing the component that survives visual degradation; HALC (Chen et al., 2024b) contrasts adaptively selected field-of-view crops instead, and Favero et al. (2024) formalise the contrast as an information-grounding objective. DeGF (Zhang et al., 2025) synthesises an image from the candidate caption and diffs it against the input. MARINE (Zhao et al., 2025a) injects object-level guidance from an external vision expert, the closest decoding-time analogue of our evidence channel — except that it steers the model rather than replacing its answer, and offers no bound. A second line localises the failure to particular heads and layers and intervenes there (Jiang et al., 2025; Wan et al., 2025; Deng & Yang, 2025; Cheng et al., 2026); a third attacks the positional prior, MCA-LLaVA (Zhao et al., 2025b) replacing RoPE's one-dimensional decay with a two-dimensional Manhattan-distance spatial decay. A fourth changes the model: factually augmented RLHF (Sun et al., 2024), RLHF-V (Yu et al., 2024), whose fine-grained *correctional* feedback is the closest training-time analogue of certified correction minus the certificate, and robust instruction tuning (Liu et al., 2024a).

All of these lower the base risk μ while we control the risk of what is emitted. Because the abstention floor is $(\mu - \alpha)/(1 - \alpha)$, *any* reduction in μ relaxes the constraint we operate under, so the two lines multiply rather than substitute; we verify this by running our layer on top of Leng et al. (2024) in §5.5. We also cite the mechanistic line as the strongest current evidence that hallucination is structured and localisable rather than noise — which is why an independent channel can be expected to be informative about it.

Mitigation by post-hoc verification and rewriting. The family our work descends from is: generate, then check, then change. Woodpecker (Yin et al., 2024) extracts object claims, queries detectors, and rewrites with a language model. LURE (Zhou et al., 2024) identifies hallucination-prone captions from co-occurrence and uncertainty statistics and revises them. REVERSE (Wu et al., 2025) trains the model to emit confidence markers and then resamples the spans it flagged, which makes it the closest work in *spirit* to ours: it also changes the output rather than deleting it. Knowledge-grounded variants replace detectors with retrieval over multimodal knowledge graphs (Lee et al., 2024) or domain corpora (Xia et al., 2024).

Every method in this family performs the same three steps we do — flag, consult external evidence, substitute — and none certifies the substitution; the quantity reported is a benchmark average after rewriting. Dang et al. (2026) is of exactly this type, and the present paper can be read as asking what it would take to attach a guarantee to such a pipeline. The attachment is not a matter of wrapping the pipeline in a calibration loop: the natural way to certify a correction is circular (§4.4), and the honest version requires an architectural commitment that this literature happens to satisfy by accident rather than by design.

2.2 Conformal prediction and distribution-free risk control

From coverage to risk. Conformal prediction (Vovk et al., 2005; Papadopoulos et al., 2002; Lei et al., 2018) converts any score into prediction sets with finite-sample marginal coverage under exchangeability alone; see Angelopoulos & Bates (2023) for a modern treatment. Refinements adapt the sets to local difficulty (Romano et al., 2019), exploit leave-one-out structure (Barber et al., 2021), control error per class (Sadinle et al., 2019; Ding et al., 2023), condition on groups (Vovk et al., 2003), and relax exchangeability itself (Barber et al., 2023). The relevant generalisation for us is from *coverage* to *risk*: risk-controlling prediction sets (Bates et al., 2021) bound the expectation of a monotone loss, conformal risk control (Angelopoulos et al., 2024) extends this to monotone risks with a single scalar parameter, and Learn-Then-Test (Angelopoulos et al., 2021) drops monotonicity altogether and treats the choice of configuration as a multiple-testing problem. The last is the tool we need: our policy family's induced risk is *not* guaranteed monotone once repairs enter the emitted set, and LTT's fixed-sequence testing lets us search a pre-ordered family at full δ without a Bonferroni penalty.

The bound matters. Which concentration inequality supplies the per-hypothesis p-value has a large practical effect at the sample sizes available in VLM calibration. Hoeffding's inequality is loosest; empirical Bernstein adapts to observed variance; betting-style e-value constructions are tighter again (Waudby-Smith & Ramdas, 2024). For a binary loss — our case, since an answer is right or wrong — the exact Clopper–Pearson interval (Clopper & Pearson, 1934) dominates all of these, and we confirm the ordering experimentally in §8. Anytime-valid inference via e-processes and Ville's inequality (Waudby-Smith & Ramdas, 2024; Ramdas et al., 2023) extends validity to data-dependent stopping, the natural machinery for sequence-level guarantees, which we take up — and find insufficient on its own — in §7.

Selective prediction is older than conformal prediction. The accept/abstain trade-off dates to Chow (1970); its statistical theory (El-Yaniv & Wiener, 2010; Bartlett & Wegkamp, 2008; Cortes et al., 2016) and its deep-learning instantiations (Geifman & El-Yaniv, 2019) predate the conformal treatment, which adds not the idea of abstaining but the distribution-free, finite-sample character of the promise. The distinction is worth keeping in view, because the impossibility bound we build on is a statement about the selective-prediction *action space* and so applies to the classical formulations too. Learning-with-rejection theory has, to our knowledge, likewise never considered a *reject-and-substitute* action with a risk certificate.

2.3 Conformal methods for language and vision–language models

Text. Conformal language modeling (Quach et al., 2024) calibrates sampling, stopping and rejection rules so that the returned set of generations contains an acceptable output with high probability. Conformal factuality (Mohri & Hashimoto, 2024) decomposes an output into sub-claims and *backs off* until a factuality guarantee holds, making the emitted object a subset of the original. Conformal alignment (Gui et al., 2024) certifies which foundation-model outputs are trustworthy enough to use; multiple-choice settings admit split conformal prediction directly (Kumar et al., 2023); and conformal machinery calibrates when an LLM planner should ask a human for help (Ren et al., 2023) — an action space of {act, defer}, again without substitution. Token-level variants calibrate on entropy or logit statistics (Xu & Lu, 2025; Chen et al., 2025), online and bootstrapped variants address feedback loops and estimator variance (Lee et al., 2025; Pang et al., 2025), and pipeline-level joint coverage has been studied (Kotte, 2026b).

Vision–language. ConfLVLM (Li et al., 2025) is the closest published system in the multimodal setting. It treats each generated detail as a hypothesis, scores it with a domain-appropriate heuristic — CLIP similarity (Radford et al., 2021) for natural-scene descriptions, BiomedCLIP for radiology, LayoutLMv3 for documents — and filters unsupported details, reporting a reduction in claim error from 87.8% to 10.0% for LLaVA-1.5 scene descriptions at a stated coverage cost. Wagner (2026) study conformal abstention for visual question answering along two axes of their own: whether the model's answer is correct and whether the question is answerable at all from the image.

The common operation. Across this entire body of work — text and multimodal, offline and online, claim-level and sequence-level — the operation performed on a flagged claim is the same: **deletion, abstention, or back-off**. This is the empirical basis for our reading of Kotte (2026a): the premise of the impossibility bound is not an artefact of that paper's formalism, it is a faithful description of what every method in the area actually does. Nothing in the literature emits a claim the model did not produce and then certifies the result.

2.4 Uncertainty quantification without a guarantee

Our channel-1 score is not novel and we do not claim it is; it belongs to a large literature on estimating generative uncertainty, which supplies the raw material that conformal wrappers calibrate. Sequence likelihood and length-normalised logprobs are the baseline; Kadavath et al. (2022) prompt models to predict their own correctness; semantic uncertainty (Kuhn et al., 2023) clusters samples by entailment before computing entropy so that paraphrases do not count as disagreement, and semantic entropy scales this into a practical hallucination detector (Farquhar et al., 2024); verbalised confidence (Lin et al., 2022) elicits a probability in words; SelfCheckGPT (Manakul et al., 2023) measures mutual consistency across samples with no logit access, and self-consistency (Wang et al., 2023b) uses the same signal to improve accuracy rather than to flag. Calibration of the underlying classifier (Guo et al., 2017) determines how much any of these are worth.

Two observations relevant to our design. First, all of these are *self-referential*: they interrogate the model about its own output. This is exactly the property that §4.4 shows to be fatal for certifying a correction — a score derived from the model cannot vouch for an answer that contradicts the model. Second, the detectors used in the mitigation literature (open-vocabulary detection (Minderer et al., 2023; Liu et al., 2024d), image–text similarity (Radford et al., 2021)) are not uncertainty estimates at all but independent perceptual evidence, and it is precisely their independence, not their accuracy, that makes them usable as channel 2.

2.5 The two closest works, and what separates us from them

The impossibility bound, and the three ways to answer it. Kotte (2026a) asks when conformal risk control can certify structured LLM outputs at all. Three results matter here. First, the impossibility result: when the base risk exceeds the target, any distribution-free method must abstain on at least $(\mu-\alpha)/(1-\alpha)$ of examples; because μ and α are known before calibration this doubles as a closed-form feasibility test. Second, a certification hierarchy across Hoeffding, empirical Bernstein and a betting-based e-CRC bound, with the Hoeffding-to-Bernstein step supplying the largest single gain (+37% certified configurations) and e-CRC earning its keep only when calibration data is scarce. Third, a validation of adaptive conformal inference under cross-dataset shift, cutting risk-target violations from 71% to 21%, with residual failures concentrated where the bound predicts. The scope is substantial — six open-weight models from 3B to 72B, eight datasets, four tasks, six nonconformity scores — and the verdict is blunt: hard NER, QA and classification configurations are uncertifiable at $\alpha = 0.10$.

Two things about it matter for us. The first is what it does *not* study: the setting is text-only structured generation, so the multimodal case, where an independent perceptual channel is available in a way it is not for pure text, is outside its scope. The second is the remedy it recommends, because it names the fork this paper takes differently. Given $\mu > \alpha$ the floor admits exactly three responses. **(i) Relax the target**, the recommendation of Kotte (2026a): at $\alpha = 0.30$–0.40 certification becomes practical (47% of NER, 40% of QA, 60% of classification configurations). **(ii) Lower μ or sharpen the score**: every mitigation method in §2.1 does the former and Xu et al. (2026) the latter; both stay inside the premise and shrink the floor without removing it. **(iii) Change the action space** — emit something the model did not say. Only for (iii) does the theorem stop applying, and it is the one we take, at the cost of re-deriving validity for the mixture (Theorem 1) and inheriting constraints the other two do not face (§4.4, Theorem 2). We do not weaken or contradict the bound; we re-derive it in our own notation in Proposition 1. The same author's pipeline-level conformal work (Kotte, 2026b) is subject to the same premise at every stage.

Budgeted conformal evidence acquisition. Xu et al. (2026) is concurrent and close enough that a reader who skims both may reasonably ask what is left. Their diagnosis is ours, stated more sharply: to hold the hallucination rate below 5% on a balanced object-existence benchmark, a state-of-the-art conformal filter must abstain on more than 80% of claims. Their remedy replaces the binary answer/abstain decision with a three-way choice — answer, abstain, or acquire additional visual evidence by re-examining the image under a bounded budget — and restores validity by calibrating on post-acquisition scores. Their setting (POPE/COCO, LLaVA and Qwen backbones), machinery (Clopper–Pearson p-values with fixed-sequence testing) and motivation coincide with ours, and in the regime we reproduce they report certified coverage rising from roughly 28% to 37% at $\alpha = 0.10$.

The distinction survives the fact that both papers describe a "three-way choice". Their third action is *acquire*: spend budget to look again, then answer or abstain on a better-informed score. Ours is *repair*: emit a different answer. BCEA changes *how confident it is* about the model's claim; we change *what claim is emitted*. So BCEA is response (ii) above and remains inside the premise of Kotte (2026a) — which is why it reduces the floor's bite without escaping it — while CCRC is response (iii). The practical consequence is that the two rescue different populations, and §5.7 finds the gains approximately additive under a common base score. We accordingly narrow our novelty claim to the substitution action and the guarantee over the resulting mixture, rather than to the observation that conformal VLM filtering abstains too much, which we share with them.

A note on the word "certified". The term is used in at least three senses in nearby literatures, and we mean the third. In certified robustness the guarantee is over a perturbation ball — deterministically for exact verifiers, and with high probability over the smoothing noise for randomized smoothing (Cohen et al., 2019). In automated program repair (Le Goues et al., 2019) a "validated" patch is one that passes a test suite, an empirical check with no statistical content. Our sense is the conformal one: a finite-sample, distribution-free, high-probability bound on the risk of the emitted mixture under exchangeability. We claim no per-item guarantee, and §9 is explicit about what the guarantee does not say.

2.6 Distribution shift caused by the predictor

Editing a generation changes the distribution of what follows it. This is not exogenous covariate shift, for which weighted conformal prediction is the standard remedy (Tibshirani et al., 2019), but *feedback* covariate shift: the shift is a function of the deployed decision rule itself (Fannjiang et al., 2022). Validity can be restored in some such settings by careful reweighting (Prinster et al., 2024), and adaptive conformal inference tracks drift online with no distributional assumptions (Gibbs & Candès, 2021). Model cascades and routing systems face a structurally similar problem (Chen et al., 2024a; Gupta et al., 2024). Our sequential experiment (§7) shows that in generative correction the obstacle is stronger than the loss of exchangeability: even granting a mechanism to restore validity under the induced shift, the intervention degrades the object being certified. Exchangeability can be repaired by reweighting; a monotonicity violation cannot.

2.7 What is new here

Positioned against the above, our claim is narrow and, we hope, correspondingly credible. Kotte (2026a) show that certifying without repairing has a hard price and Xu et al. (2026) show how to reduce — but not escape — that price from inside the premise. What is new here is a selective predictor whose action space includes substitution and whose guarantee covers the resulting mixture, together with the four measurements that bound it: the budget on self-certified substitution, the decomposition of where the gain actually comes from, the small-calibration failure mode of the sequential test, and the two measured impossibilities (symmetric-gate repair of false positives, and sequence-level monotonicity). The negative half is the more durable one.

Table 1: Notation. Symbols are used with these meanings throughout; where a letter is reused in a different role we mark it explicitly at the point of use.

symbol	meaning
α	risk target for the emitted set
δ	failure probability of the certificate
μ	base risk of the underlying model, $\Pr[Y = 0]$
$a^\star$, $\hat{a}(X)$	gold answer; the model's answer
Y	correctness *indicator* of the model's answer, $\mathbf{1}\{\hat{a} = a^\star\}$
$Y^{(2)}$	correctness indicator of channel 2's answer
$s(X)$	channel-1 correctness (conformity) score; large means likely correct
$m(X)$	channel-2 evidence margin
$t \in \mathcal{T}$	acceptance threshold *value*, from a grid $t_1 > \cdots > t_M$ fixed before calibration
g	*fixed* repair-gate threshold value on m, never calibrated
λ, q	permissiveness and repair quantile of the rank-indexed variant we implement, with $t = Q_s(1-\lambda)$ and $g = Q_m(1-q)$ (Remark 3)
$Q_s(\cdot), Q_m(\cdot)$	empirical quantiles of s, m on the calibration fold
K_t, V_t	emitted count; error count among emitted (written K_λ, V_λ in the rank-indexed variant)
$\mathcal{R}, \mathcal{C}$	risk of, and coverage of, the emitted set
$k_{\min}$	smallest emitted count that can be certified (Lemma 1)
$\mathcal{I}$	sequential interference term (Theorem 2)

3 Problem setup

3.1 Selective prediction with a risk guarantee

Table 1 fixes notation. Two conventions are worth stating because nearby literatures differ: α is always the risk *target* and never a significance level, and δ is always the failure probability of the certificate.

Let $(X, a^\star)$ be a query–answer pair, where X contains an image and a question. A VLM produces $\hat{a}(X)$. Define the model's correctness indicator $Y = \mathbf{1}\{\hat{a}(X) = a^\star\}$ and the *base risk* $\mu = \Pr[Y = 0]$. A selective predictor is a measurable rule that either emits an answer or abstains. Writing $\mathcal{E}$ for the (random) set of emitted items,

$$\mathcal{R}(\mathcal{E}) = \Pr[\text{emitted answer wrong} \mid \mathcal{E}], \qquad \mathcal{C}(\mathcal{E}) = \Pr[\text{item emitted}].$$

The goal is a policy with $\mathcal{R}(\mathcal{E}) \leq \alpha$ certified at confidence $1 - \delta$, and $\mathcal{C}(\mathcal{E})$ as large as possible. We assume throughout:

Assumption 1 (Exchangeability)**.** Calibration and test items are exchangeable draws from the same distribution.

Assumption 2 (Pre-specified policy family)**.** The candidate policies form a family $\{\pi_t\}_{t \in \mathcal{T}}$ whose index set and ordering are fixed before seeing calibration data. In our formulation $\mathcal{T}$ is a grid of *score values*, not of ranks or quantiles; Remark 3 discusses the difference, which is the difference between a two-line proof and an open problem.

3.2 Filtering, and the floor it cannot cross

A *filtering* policy uses a score $s : \mathcal{X} \to [0, 1]$, intended to increase with correctness, and emits $\hat{a}(X)$ when $s(X) \geq t$.

Proposition 1 (Abstention floor; Kotte 2026a, Prop. 3)**.** *Let $\mu > \alpha$. Any selective predictor whose emitted answer is always $\hat{a}(X)$ satisfies*

$$1 - \mathcal{C}(\mathcal{E}) \;\geq\; \frac{\mu - \alpha}{1 - \alpha}$$

whenever $\mathcal{R}(\mathcal{E}) \leq \alpha$. The bound is independent of the score s, of the concentration inequality used, and of the calibration sample size.

Table 2: The abstention floor against the abstention actually incurred by our filtering pipeline at $\alpha = 0.10$ (Table 9). The floor is real and irreducible, and it is a minority of the bill.

Setting	μ	floor $\frac{\mu-\alpha}{1-\alpha}$	certified cov.	abstention	floor share
POPE-1500 LLaVA	18.1%	9.0%	68.6%	31.4%	29%
POPE-adv LLaVA	17.3%	8.1%	44.0%	56.0%	14%
POPE-adv Qwen2-VL	12.9%	3.2%	77.9%	22.1%	15%
POPE-adv LLaVA+VCD	19.6%	10.7%	48.8%	51.2%	21%
AMBER(d) LLaVA	11.4%	1.6%	91.8%	8.2%	20%

Proof in our notation. Let $c = \mathcal{C}(\mathcal{E}) = \Pr[X \in \mathcal{E}]$ and split the total error mass by emission:

$$\mu \;=\; \Pr[Y=0] \;=\; \underbrace{\Pr[Y=0 \mid X \in \mathcal{E}]}_{=\,\mathcal{R}(\mathcal{E})}\, c \;+\; \Pr[Y=0 \mid X \notin \mathcal{E}]\,(1-c).$$

The second conditional probability is at most 1, and the emitted answer is always $\hat{a}(X)$, so no rule can make an abstained item's model answer correct — abstention removes the item from the emitted set but does not change Y. Hence $\mu \le \mathcal{R}(\mathcal{E})\, c + (1-c)$. Imposing $\mathcal{R}(\mathcal{E}) \le \alpha$ gives $\mu \le \alpha c + 1 - c = 1 - c(1-\alpha)$, i.e. $c \le (1-\mu)/(1-\alpha)$ and therefore

$$1 - c \;\ge\; 1 - \frac{1-\mu}{1-\alpha} \;=\; \frac{\mu-\alpha}{1-\alpha}.$$

Nothing in the argument refers to s, to the concentration inequality, or to the calibration size, which is the sense in which the bound cannot be engineered away. The single place it uses the premise is the step "no rule can make an abstained item's model answer correct": it is exactly this step that fails when the predictor may emit an answer other than $\hat{a}(X)$, and §4 is the consequence. $\square$

Two features of Proposition 1 drive this paper. It is *computable in advance*, since μ and α are known before any calibration; and it is *conditional on a premise* — that the emitted answer is the model's own — which is an assumption about the action space, not about the data.

3.3 How much of the abstention is the floor? (Not much.)

Before building on the floor it is worth measuring how much of the bill it actually explains, because a reader who sees "at least 9.0% must be refused" next to "68.6% certified coverage" can easily attribute the whole 31.4-point gap to it. Table 2 does the arithmetic. The floor accounts for 14–29% of realised abstention on our five settings at $\alpha = 0.10$; on POPE-adversarial with LLaVA it is 8.1 points out of 56.0.

The remaining three-quarters is score quality and finite-sample power. Two consequences follow, and both are worth stating plainly. First, the floor is the right reason to look at the *action space* and the wrong reason to expect large gains from doing so: it is the part of the bill that no score and no sample size can retire, which is why substitution is worth introducing, but retiring it recovers a fraction of what is being paid. Second, the honest ordering of magnitudes in this paper is score $\gg$ power $>$ action space: 64.5 points from the correctness score against 2.3 from certified correction (§5.6), and a failure mode driven purely by calibration-fold size (§6) that costs more than our mechanism gains. We report the 2.3.

3.4 Two channels

Definition 1 (Channels)**.** *Channel 1* is a correctness score $s(X) = \widehat{\Pr}[Y = 1 \mid \phi(X)]$ fitted on features $\phi(X)$ derived from the model's own output distribution together with grounding features. Note the orientation: s is a *conformity* score, large for items the model is likely to get right, and acceptance is an upper-tail event $s \ge Q_s(1-\lambda)$. This is the mirror image of the usual nonconformity convention; we use it because it reads naturally as "confidence that the answer is correct". *Channel 2* is a separate predictor $b(X)$ that answers the query independently of the VLM, together with an evidence margin $m(X) \in [0,1]$ measuring how decisively it does so. We write $Y^{(2)} = \mathbf{1}\{b(X) = a^\star\}$ for channel 2's correctness. The only property the analysis uses is that $Y^{(2)}$ is not a deterministic function of Y; we do not assume statistical independence, and §9 notes that our instantiation violates it (channel 1's features include the detector score).

Remark 1 (Our instantiation repairs in one direction only). This is a property of our channel 2, not of CCRC, but it bounds every positive result below and we state it before the experiments rather than after. We instantiate b by thresholding an open-vocabulary detector's confidence $o(X)$ at $\theta = 0.15$ and set $m(X) = |o(X) - \theta|$. A maximally confident detector *negative* therefore attains only $m \le \theta = 0.15$, whereas positives reach 0.85. The gate $g = Q_m(1-q)$ at $q = 0.10$ sits at 0.383, 0.357, 0.366, 0.366 and 0.209 on our five settings — above θ in every case. Consequently 100.0% **of the repaired mass is "detector says the object is present"** in all ten (setting, α) cells, measured over 500 paired splits (§5.3, Table 10), and CCRC as evaluated can repair only *missed* objects. It never repairs the canonical false-positive existence hallucination that POPE-adversarial is built to elicit; those items are abstained, not corrected.

It also explains why the repaired mass is small. Stating the denominator, since it is easy to mistake: repaired mass is the number of repaired items as a fraction of *the test fold*, which is $n/3$ items under the three-way split, and it is 0.97–1.97% at $\alpha = 0.10$ and 0.35–1.36% at $\alpha = 0.15$ — so on POPE-1500 (test fold 500) a typical split repairs about nine items. All "repaired mass" percentages in this paper use that denominator, and all gains are percentage points of the same test fold, which is what makes the two directly comparable in §4.7.

A symmetric margin with polarity-specific gates is the obvious remedy. We have now evaluated it: it is valid, it admits negative-polarity repairs at 99.9–100% precision, and it repairs no false positives at all (§5.3). The restriction above is therefore not an artefact of our margin's parameterisation.

In our instantiation channel 1 combines the VLM's yes/no probability, its confidence margin, and grounding features; channel 2 is an open-vocabulary object detector (Minderer et al., 2023) queried with the object named in the question, with m the absolute margin of its detection score around a fixed operating point. The essential property is that $Y^{(2)}$ is *not* a deterministic function of Y — the two channels can be right and wrong independently. §4.4 shows this is what makes repair possible.

4 Certified-correction risk control

4.1 The procedure

Fix a strict evidence gate g and a descending grid of acceptance thresholds $t_1 > t_2 > \cdots > t_M$, both before seeing calibration data. For $t \in \mathcal{T} = \{t_j\}$, policy π_t acts as

$$\pi_t(x) = \begin{cases} \textsc{accept } \hat{a}(x), & s(x) \ge t, \\ \textsc{repair with } b(x), & s(x) < t \text{ and } m(x) \ge g, \\ \textsc{abstain}, & \text{otherwise.} \end{cases} \tag{1}$$

Figure 1 shows the induced partition of the (s, m) plane. The emitted set is $\mathcal{E}_t = A_t \cup R_t$ with A_t the accepted and R_t the repaired items, and the emitted-set error counts what is *actually emitted*,

$$V_t = \sum_{i \in A_t} (1 - Y_i) + \sum_{i \in R_t} \big(1 - Y_i^{(2)}\big), \qquad K_t = |A_t| + |R_t|. \tag{2}$$

Both thresholds are values on the score axes rather than ranks in the calibration sample. That choice is what makes K_t a random variable and the emitted error indicators conditionally i.i.d., which in turn makes Theorem 1 two lines long; it is worth the paragraph we spend on it in Remark 3. Our implementation sets the two values from calibration-fold quantiles, $g = Q_m(1 - q)$ with the fixed $q = 0.10$ (repairs come only from the top decile of channel-2 evidence) and $t_j = Q_s(1 - \lambda_j)$ on a grid of permissiveness $\lambda_1 < \cdots < \lambda_M$; that variant is the one we measure everywhere, and Remark 3 reports what we know about its validity.

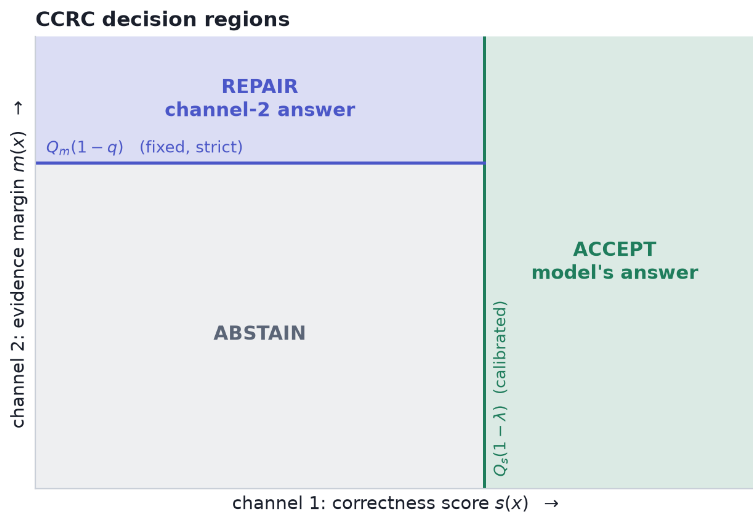

Figure 1: CCRC partitions the (s, m) plane. The acceptance threshold $Q_s(1-\lambda)$ is calibrated; the repair gate $Q_m(1-q)$ is fixed in advance and strict, so that as λ loosens the accepted region grows and the repair region shrinks. Keeping q fixed is what makes the family nested and lets a single fixed-sequence test suffice (§4.5).

4.2 Worked examples on real data

Before turning to aggregates, Figure 2 shows what the three decisions look like on individual POPE images, with the actual channel-1 score, channel-2 detection confidence, and resulting action for each. The three columns illustrate the regimes the procedure is designed to separate.

In the ACCEPT column the model answers a straightforward existence question confidently and correctly, the detector agrees, and the item is emitted unchanged; these cases carry the bulk of the coverage. The REPAIR column contains the cases that motivate the method: LLaVA is asked whether a computer mouse is present, answers "no" with $p = 0.21$, and is *wrong* — the mouse is there. Channel 1 correctly assigns a low correctness score ($s = 0.18$), which under a filtering policy would produce an abstention and no answer at all. The detector, however, localises the object with confidence 0.90, comfortably inside the top evidence decile, so CCRC substitutes the detector's answer and the item is emitted *correctly*. This is the dilution effect of §4.7 in a single instance: a high-precision repair converts an item that filtering would discard into a low-risk emitted item.

The ABSTAIN column shows the necessary complement. Here the model is wrong *and* the detector is also wrong, with weak evidence in both cases ($m = 0.04$ and $m = 0.03$); the questions ask about a handbag and a truck that are not present, and neither channel supplies a certifiable answer. Abstention is the correct action, and no amount of substitution would help. The existence of a substantial population of such items is why CCRC's gains are bounded by Proposition 4 rather than being able to recover the entire abstained mass.

4.3 Validity

Theorem 1 (CCRC validity)**.** *Let Assumptions 1–2 hold, let $\mathcal{T} = \{t_1 > \cdots > t_M\}$ and the repair gate g be fixed before calibration, and let*

$$U(e, k; \delta) = \sup\{p : \Pr[\mathrm{Bin}(k, p) \leq e] > \delta\}$$

be the one-sided Clopper–Pearson upper confidence bound. Define $\hat{t}$ by fixed-sequence testing: proceed through $j = 1, 2, \ldots$ while $U(V_{t_j}, K_{t_j}; \delta) \leq \alpha$ on the calibration fold, and set $\hat{t}$ to the last index that passed. Then the

Figure 2: Real POPE images with the actual quantities CCRC uses. Each panel reports the question, the ground truth, LLaVA's answer with its probability, the detector's answer with its confidence and evidence margin, the channel-1 correctness score, and the resulting decision. **Accept** (left): both channels agree and the model is right. **Repair** (centre): the model misses an object that is genuinely present and receives a low correctness score, while the detector localises it with high confidence, so the answer is substituted and the error is fixed — items a filtering policy would abstain on. **Abstain** (right): the model is wrong and the detector is also wrong with weak evidence, so no certifiable answer exists and the procedure correctly declines. Thresholds are the calibrated $Q_s(1-\lambda)$ and the fixed gate $Q_m(1-q)$ from one split.

deployed policy $\pi_{\hat{t}}$ satisfies

$$\Pr\big[\mathcal{R}(\mathcal{E}_{\hat{t}}) \le \alpha\big] \;\ge\; 1-\delta.$$

Proof. Fix j and consider the null $H_j : \mathcal{R}(\mathcal{E}_{t_j}) > \alpha$. Because t_j and g are constants, membership in the emitted set, $\{s_i \ge t_j\} \cup \{s_i < t_j,\ m_i \ge g\}$, is a function of item i alone, and each emitted item contributes exactly one error indicator: $1 - Y_i$ if accepted, $1 - Y_i^{(2)}$ if repaired. Under Assumption 1 the items are exchangeable draws, so conditional on $K_{t_j} = k$ the k emitted indicators are i.i.d. Bernoulli with common mean $\mathcal{R}(\mathcal{E}_{t_j})$ — selection is by a set fixed in advance, which does not tilt the conditional error probability, and k is random but ancillary. Hence $V_{t_j} \mid K_{t_j} = k \sim \mathrm{Bin}\big(k, \mathcal{R}(\mathcal{E}_{t_j})\big)$ and, by construction of the Clopper–Pearson bound, $\Pr[U(V_{t_j}, k; \delta) \le \alpha \mid H_j] \le \delta$: the event that the bound certifies a policy whose true risk exceeds α has probability at most δ, so $\phi_j = \mathbf{1}\{U \le \alpha\}$ is a valid level-δ test of H_j.

The index set and its ordering are pre-specified by Assumption 2, which is all that fixed-sequence testing requires: testing $H_1, H_2, \dots$ in that order and stopping at the first non-rejection controls the family-wise error rate at δ without any multiplicity correction (Angelopoulos et al., 2021). The returned $\hat{t}$ is among the rejected hypotheses, so with probability at least $1-\delta$ every rejected H_j is false, i.e. $\mathcal{R}(\mathcal{E}_{\hat{t}}) \le \alpha$. □

Two structural remarks follow; the second is the one a careful reader should hold us to.

Remark 2 (why the mixture is not a complication, and why nesting is not a hypothesis). The proof needs only that each emitted item contributes one $\{0, 1\}$ error indicator and that the set it is drawn from is fixed in advance. Whether that indicator refers to the model's answer or to channel 2's is irrelevant, which is precisely why Proposition 1 does not bind: its premise restricts the *action space*, which the guarantee itself never uses. Nesting is likewise not a validity hypothesis, but it is what makes the procedure powerful: because g is fixed, A_t grows and R_t shrinks as t falls, so $\mathcal{E}_t = A_t \cup M$ with $M = \{m \geq g\}$ constant, and *coverage* increases along the grid even though the risk need not — which is what makes "the last index that passed" the right thing to return.

Remark 3 (value-indexed versus rank-indexed families). The natural alternative indexes the family by quantiles, $t = Q_s(1-\lambda)$ with Q_s the empirical quantile of the *same* calibration fold whose errors are counted, and would like the emitted-error count to be binomial with mean $k\,\mathcal{R}$. That step is where the difficulty actually lives, and it does not hold: selecting the top $\lceil \lambda n_{\mathrm{cal}} \rceil$ items by score makes k deterministic and makes the selection event a function of all items jointly, so conditionally on the calibration features the emitted indicators have item-specific means and V is *Poisson binomial* rather than binomial. Poisson-binomial lower tails are dominated by the equal-mean binomial (Hoeffding, 1956), which points in the conservative direction, but that domination is not by itself a treatment of the order-statistic selection effect. Indexing by score *values* removes the problem instead of bounding it, at the price of one extra pre-specification, and it costs nothing in practice: a value grid may be laid down on a held-out score histogram, as Algorithm 1 does.

We nonetheless report measurements from the rank-indexed implementation throughout, because it is what our scripts run and what a practitioner without a pre-specified value grid would write. Two pieces of evidence bound the exposure. First, a synthetic study in which the population risk of the selected policy is known exactly: over four score shapes (well-separated logistic, weakly separated, heavy-tailed, and a miscalibrated monotone transform) × four calibration sizes (100, 250, 500, 1500), the realised exceedance $\Pr[\mathcal{R}(\mathcal{E}_{\hat{\lambda}}) > \alpha]$ of the rank-indexed procedure is 0.0–5.9% against $\delta = 0.10$ — conservative in all sixteen cells, and nowhere near the nominal level. Second, on real data the population-risk exceedance is 0.0–5.1% across all 40 cells of §5 (Table 20). We therefore claim the theorem for the value-indexed family, and for the rank-indexed variant we claim empirical validity with a stated margin, not a proof.

One methodological note on that simulation, because it bears on how the audit tables in this paper should be read. The guarantee is unconditional, and an aborted run emits nothing and so cannot violate it; the 0.0–5.9% above therefore counts aborted draws as compliant. Conditional on the procedure certifying *something*, exceedance in the same simulation reaches much higher values on the cells where aborts dominate — up to 100% in a cell whose abort rate is also 100%, which is a statement about a handful of lucky draws and not about validity. Our real-data audit uses the conditional convention (Table 20 excludes aborted splits and prints the abort rate beside each figure), which is the conservative choice; it stays below δ everywhere on real data only because the real abort rates in that table are small.

Lemma 1 (Minimum certifiable emitted-set size)**.** *With zero observed errors, $U(0,k;\delta) = 1 - \delta^{1/k}$. Hence no emitted set of size k can be certified at level α unless*

$$k \;\geq\; k_{\min} = \frac{\ln \delta}{\ln(1-\alpha)}.$$

For $\alpha = \delta = 0.10$, $k_{\min} = \lceil 21.85 \rceil = 22$.

Proof. $\Pr[\mathrm{Bin}(k,p) \leq 0] = (1-p)^k$, so the upper bound with $e = 0$ solves $(1-p)^k = \delta$, giving $U(0,k;\delta) = 1 - \delta^{1/k}$. This is decreasing in k, and $1 - \delta^{1/k} \leq \alpha \iff \delta^{1/k} \geq 1 - \alpha \iff k^{-1} \ln \delta \geq \ln(1-\alpha)$; dividing by the negative quantity $\ln(1-\alpha)$ reverses the inequality and yields $k \geq \ln \delta / \ln(1-\alpha)$. Since $U(e,k;\delta) \geq U(0,k;\delta)$ for $e \geq 0$, no error count can do better. □

Lemma 1 is elementary but consequential: a fixed sequence whose first policy emits fewer than $k_{\min}$ items fails at step one and, because fixed-sequence testing stops at the first non-rejection, returns *zero* coverage. We encountered exactly this failure and now begin the grid at the analytically derived minimum. It turns out to be the single most consequential fact in the paper's empirical half: §6 shows that the same mechanism, one step further on, explains a weak score certifying nothing rather than less, the collapse of self-repair, and our one negative result.

4.4 Why repairs cannot be self-certified in practice

The most economical design would avoid a second channel: if s says the model is probably wrong, emit the opposite. This section shows why that fails. The argument has to be made at the level of the *mixture*, because that is what CCRC certifies — a repair region whose own risk exceeds α is not automatically inadmissible, since a compliant blend can absorb it (Proposition 3). The correct statement is therefore not that self-repair is impossible but that it is *self-defeating*: the mass it can buy is bounded by the accepted region's slack, and the fixed-sequence test aborts before it can be collected.

Proposition 2 (Self-repair is bounded by slack, and is not free)**.** *Suppose the answer space is binary and the repair is the negation of the model's answer, so that $Y^{(2)} = 1 - Y$. Let $R \subseteq \mathcal{X}$ be any repair region with mass c_r and let the accepted region have mass c_a and risk r_a. Then the risk of the repaired region is*

$$r_r \;=\; \Pr[\textit{repaired answer wrong} \mid X \in R] \;=\; \Pr[Y = 1 \mid X \in R],$$

*i.e. exactly the model's **accuracy** on R. Consequently, if $r_r > \alpha$ the mixture is compliant only for*

$$c_r \;\le\; c_a \,\frac{\alpha - r_a}{r_r - \alpha}, \tag{3}$$

and a region of accuracy $\le \alpha$ — for which no such constraint binds — requires the score to isolate inputs on which the model is right less than α of the time. That is strictly stronger than isolating inputs of low confidence*, and no monotone transformation of s can supply the difference.*

Proof. On R the emitted answer is $1 - \hat{a}$, which is correct exactly when $\hat{a}$ is incorrect; its error indicator is therefore Y, and averaging over R gives $r_r = \Pr[Y = 1 \mid X \in R]$. For the mixture, $\bar{r} = (c_a r_a + c_r r_r)/(c_a + c_r) \le \alpha$ rearranges to $c_r(r_r - \alpha) \le c_a(\alpha - r_a)$, which is (3) when $r_r > \alpha$ and is vacuous when $r_r \le \alpha$. The final claim follows because any repair rule of the form $\{s(X) \le t\}$ has $r_r = \Pr[Y = 1 \mid s(X) \le t]$, a functional of the joint law of (s, Y) invariant to monotone reparameterisation of s. □

Proposition 2 converts an intuition — "a hallucination score reports suspicion, not knowledge of the truth" — into a budget. Two measurements show how little that budget buys.

No low-accuracy region exists where it is needed. The model's accuracy in the bottom 2% of s — which is exactly r_r, the flip region's own risk — is 22.1% on POPE-1500 and 43.8% on POPE-adversarial against $\alpha = 0.10$; in the bottom decile it is 42.8% and 47.3%. On Qwen2-VL the bottom 2% is *above* chance at 62.4%, so flipping there is actively harmful. Only two settings contain a flippable region at all, VCD-decoded LLaVA at 4.9% and AMBER at 0.9%, and they are the two settings whose *accepted region has the least slack to spend*: AMBER already certifies 91.8% of its inputs at $\alpha = 0.10$ and VCD's accepted risk sits close to the budget. (This is a statement about slack in (3), not about headroom $\mu - \alpha$: VCD in fact has the *largest* headroom of our five settings, 9.6 points, and AMBER the smallest, 1.4; the two quantities are easy to conflate and point in opposite directions here.)

The fixed sequence aborts before the slack can be spent. Substituting the negation of the model's answer at $q = 0.02$ and certifying the mixture exactly as in §4.3, certified coverage falls by 49.3, 37.3, 76.9 and 6.6 points on the four POPE settings at $\alpha = 0.10$ (all $p < 10^{-7}$; Table 3). The mechanism is not a graceful trade: the fixed sequence returns *nothing at all* on 73.0%, 88.8%, 98.8% and 25.2% of splits respectively. The reason is structural. The self-repair region has fixed mass, so at the *start* of the grid — where the accepted set is smallest and K barely clears $k_{\min}$ — it is the largest fraction of the emitted set it will ever be, and its 4.9–62.4% error rate fails the very first hypothesis. Fixed-sequence testing must stop at the first non-rejection (§4.3), so the procedure never reaches the later thresholds at which (3) would in fact be satisfiable. Self-repair is defeated by the interaction between a fixed-mass, high-error repair block and the prefix structure of the test, not by the region being uncertifiable in isolation. This is the first of the three instances of that interaction collected in §6.

Table 3: Self-certified repair: flipping the model's answer on the bottom $q = 0.02$ of s and certifying the mixture, against filtering, under the identical protocol (400 paired splits, $\delta = 0.10$). r_r is the flip region's own risk, i.e. the model's accuracy there. All p-values are two-sided paired t-tests on the 400 paired differences. The gain is negative wherever the abort rate is large, and positive only where r_r is small.

Setting ($\alpha = 0.10$)	r_r	Filter	Self-repair	Gain [95% CI]	p	abort
POPE-1500 LLaVA	22.1%	68.6%	19.3%	-49.3 $[-52.5, -46.1]$	4×10^{-106}	73.0%
POPE-adv LLaVA	43.8%	44.0%	6.6%	-37.3 $[-40.1, -34.5]$	2×10^{-88}	88.8%
POPE-adv Qwen2-VL	62.4%	77.9%	1.1%	-76.9 $[-78.6, -75.1]$	1×10^{-261}	98.8%
POPE-adv LLaVA+VCD	4.9%	48.8%	42.2%	-6.6 $[-8.9, -4.3]$	4×10^{-8}	25.2%
AMBER(d) LLaVA	0.9%	91.8%	92.7%	$+0.9$ $[-0.6, +2.5]$	0.25 (n.s.)	2.8%
at $\alpha = 0.15$, the two clean-flip-region cells become positive:						
POPE-adv LLaVA+VCD	4.9%	75.8%	77.7%	$+1.9$ $[+0.6, +3.3]$	0.006	2.5%
AMBER(d) LLaVA	0.9%	95.9%	98.7%	$+2.9$ $[+2.6, +3.1]$	6×10^{-78}	0.0%

Table 4: Channel-2 (detector) accuracy by evidence-margin decile on POPE-1500 (150 items per decile). Repairs are drawn only from the top decile, where accuracy is 100%, far above $1 - \alpha$. Accuracy improves with the margin but not monotonically — the 5th falls below the 4th and the 8th below the 7th — which is why the repair gate must be strict rather than merely above the median. All ten deciles are printed because the non-monotonicity is the point.

Margin decile	1 (bottom)	2	3	4	5	6	7	8	9	10 (top)
Detector accuracy	0.513	0.713	0.733	0.807	0.772	0.934	0.953	0.927	0.960	1.000

What this licenses us to claim. Not that certified correction is impossible without a second channel. Equation (3) admits self-repair whenever the accepted region has slack and the flip region is clean enough, and that is confirmed rather than refuted by the two cells where both conditions hold: at $\alpha = 0.15$ self-repair gains +2.9 points on AMBER ($p = 6 \times 10^{-78}$) and +1.9 on VCD ($p = 0.006$). At $\alpha = 0.10$ the AMBER cell is +0.9 points with 95% CI $[-0.6, +2.5]$, $p = 0.25$: not distinguishable from zero, and we do not count it. What the measurements license is the weaker and sufficient claim that *on the settings where filtering is expensive enough for correction to be worth doing, no self-certifiable flip region exists*, so a second channel is required in practice. We state this as an empirical regularity with a mechanism, not as a theorem, and §9 lists it among the claims most likely to fail on a new benchmark.

Channel 2 escapes the budget in (3) because $Y^{(2)}$ is a genuinely different random variable, not a deterministic function of Y. The top evidence decile, which is the only decile repairs are drawn from, is 100% accurate on POPE (Table 4), so r_r there is far below α and the constraint is vacuous rather than binding.

4.5 Why the repair gate must be decoupled

The obvious first version of the procedure ties the repair gate to λ, i.e. uses $m(x) \geq Q_m(1-\lambda)$ in (1). This is natural but harmful: as λ loosens to admit more accepted items, the repair gate loosens in step and begins admitting low-precision repairs, so the fixed sequence terminates earlier. Empirically the coupled version gained on LLaVA (+4.7, +5.3 points) but *lost* on Qwen2-VL (-4.2) and on a VCD-decoded model (-2.9). It also required certifying two families (with and without repair) at $\delta/2$ and selecting between them, wasting half the confidence budget.

Fixing q removes both problems. The family becomes nested in the single threshold t, so one fixed-sequence test at full δ suffices (Theorem 1), and repair precision is held constant while t sweeps. Table 5 reports the sensitivity over all ten cells.

Table 5: Sensitivity to the repair gate q: change in certified coverage relative to filtering, over five settings × two risk levels (400 paired splits). Bracketed figures are 95% CIs on the mean paired gain. The two negative cells at $q = 0.10$ are the two AMBER cells; restricting the table to the eight POPE cells would exclude the loss, so all ten are shown.

Setting	α	$q = 0.10$	$q = 0.25$	$q = 0.50$
POPE-1500 LLaVA	0.10	+3.18 [+2.6, +3.8]	+5.39 [+3.9, +6.9]	+8.84 [+7.7, +10.0]
POPE-adv LLaVA	0.10	+3.19 [+1.7, +4.7]	−0.05 [−2.9, +2.8]	+2.04 [−1.4, +5.5]
POPE-adv Qwen2-VL	0.10	+1.60 [+0.8, +2.4]	−16.44 [−20.0, −12.9]	−14.86 [−18.3, −11.4]
POPE-adv LLaVA+VCD	0.10	+2.15 [+1.3, +3.0]	−6.35 [−9.2, −3.5]	−9.04 [−12.0, −6.1]
AMBER(d) LLaVA	0.10	−10.20 [−13.4, −7.0]	−38.69 [−43.3, −34.1]	−25.91 [−30.2, −21.6]
POPE-1500 LLaVA	0.15	+1.95 [+1.8, +2.1]	+5.38 [+5.2, +5.6]	+6.78 [+6.6, +7.0]
POPE-adv LLaVA	0.15	+7.59 [+5.5, +9.7]	+10.14 [+7.7, +12.6]	+14.47 [+12.0, +16.9]
POPE-adv Qwen2-VL	0.15	+0.40 [+0.3, +0.5]	−1.22 [−2.6, +0.2]	+0.28 [−0.7, +1.3]
POPE-adv LLaVA+VCD	0.15	+1.39 [+0.5, +2.3]	+0.48 [−1.0, +2.0]	+1.40 [+0.1, +2.7]
AMBER(d) LLaVA	0.15	−11.90 [−15.1, −8.7]	−11.55 [−14.7, −8.4]	−2.19 [−3.8, −0.6]
cells positive		**8 / 10**	5 / 10	6 / 10
range		−11.90 to +7.59	−38.69 to +10.14	−25.91 to +14.47
worst abort rate		17.5%	43.8%	36.0%

4.6 What "q is fixed a priori" can and cannot mean

Asserting that q is fixed in advance and not tuned would not be defensible next to a table showing $q = 0.10$ to be the best of three values on the same cells that produce the headline. Table 5 makes the honest version available. Three statements, in decreasing order of comfort.

First, $q = 0.10$ **is not the best value by gain.** At $\alpha = 0.15$, $q = 0.50$ wins all five cells, and it also wins POPE-1500 at $\alpha = 0.10$; the mean gain over the eight POPE cells is +2.66 at $q = 0.10$ against +4.89 for a per-cell oracle. What $q = 0.10$ is best at is *robustness*: it is positive in 8 of 10 cells against 5 and 6, its worst cell is three times less bad, and its worst abort rate is less than half. That is a legitimate reason to prefer a strict gate and it is the reason we do.

Second, **the selection exposure is small and points the wrong way for us.** Choosing q by leave-one-dataset-out — maximising mean gain on the other datasets at the deployment α, which is user-specified rather than tuned — yields a mean gain of +4.19 points over the eight POPE cells against the +2.66 we report for fixed $q = 0.10$, with an oracle–LODO gap (the actual selection exposure) of +0.70 points. A properly transferred choice of q would therefore make this paper's numbers *larger*; the reported configuration is conservative.

Third, q **must be indexed to** α**.** Conditional on α, the LODO choice is stable ($q = 0.10$ at $\alpha = 0.10$, $q = 0.50$ at $\alpha = 0.15$, on every fold). Pooling across α destroys it: the pooled rule picks $q = 0.50$ for the Qwen $\alpha = 0.10$ fold and turns +1.31 into −14.71. Anyone transporting our recommendation to a different risk target should re-select, not reuse. Appendix A.4 reports a fully valid alternative that removes the objection outright — certify all three q at $\delta/3$ and select on the calibration fold — at a mean cost of +0.50 points relative to the reported configuration.

4.7 Why it gains: dilution, and how much of the gain it explains

Our initial hypothesis was that repairs *spend* the accepted set's unused risk budget: filtering typically realises a risk below α (we measure 8.0% at $\alpha = 0.10$), and repairs with error above α could be admitted while the blend stayed compliant. The measured effect runs the other way, and is stronger.

Proposition 3 (Dilution lowers the emitted risk pointwise in λ)**.** *Fix λ and let the accepted set have mass c_a and risk r_a, and the repair set mass c_r and risk r_r. The emitted risk is the mixture $\bar{r} = (c_a r_a + c_r r_r)/(c_a + c_r)$. If $r_r < r_a$ then $\bar{r} < r_a$, strictly decreasing in c_r, so every λ that is population-feasible for filtering remains* *population-feasible with repairs and further λ may become feasible. If instead $r_r > \alpha > r_a$, admitting repairs consumes slack and may render feasible λ infeasible.*

Proof. $\bar{r} - r_a = c_r(r_r - r_a)/(c_a + c_r)$, which is negative iff $r_r < r_a$, and $\partial\bar{r}/\partial c_r = c_a(r_r - r_a)/(c_a + c_r)^2$ has the same sign. Filtering is the special case $c_r = 0$, giving the feasibility claim. The last statement is the same computation with the inequality reversed. ☐

Remark 4 (Why this does not immediately imply a more permissive $\hat{\lambda}$). Proposition 3 is a statement about the *population* risk at fixed λ, and two gaps separate it from the certified $\hat{\lambda}$ the procedure returns. First, $\hat{\lambda}$ is chosen by a Clopper–Pearson bound on realised counts, not by population feasibility: a single repaired item that happens to be wrong raises V_λ and can tighten $\hat{\lambda}$ even when $r_r < r_a$. Second, fixed-sequence testing stops at the first non-rejection, so a larger feasible *set* yields a later stopping index only if feasibility is a *prefix* of the grid — which is risk monotonicity in λ, exactly what §4.3 declines to assume. We therefore treat the coverage gain as an empirical claim (§5.2) for which Proposition 3 supplies the mechanism, not a corollary. §4.4 exhibits the failure mode this remark anticipates: there, feasibility is not a prefix and the test aborts at the first index.

The empirically relevant regime is the first. Because repairs are drawn from the top evidence decile, their risk is *lower* than that of the acceptance gate's *marginal* items — the items that determine whether the constraint binds. Repairs therefore act as ballast: a small high-precision mass pulls $\bar{r}$ down and can let the threshold open further.

How much of the gain is that, and how much is arithmetic. One tempting way to report the effect is a "leverage multiplier": the small repaired mass buying more coverage than its own size. That framing is largely tautological and we do not use it. Repaired items are *themselves emitted*, so whenever $\hat{\lambda}$ does not fall, CCRC emits exactly filtering's accepted set plus the gated repairs and the gain is mechanically equal to the repaired mass — no dilution required, and nothing about the mechanism demonstrated. The quantity that does demonstrate it is the *extra accepted* mass admitted because $\hat{\lambda}$ actually moved. So we decompose the gain per split into

$$\underbrace{\text{repairs available at filtering's own } \hat{\lambda}}_{\text{(i) automatic}} + \underbrace{\text{everything else}}_{\text{(ii) dilution}},$$

computed on the splits where both arms certify something, and report both (Table 6).

Two facts govern the reading. $\hat{\lambda}$ is *identical* between filtering and CCRC on 70.6–87.7% of splits at $\alpha = 0.10$ and 76.5–98.3% at $\alpha = 0.15$; on those splits there is no dilution effect at all, by construction. And where it does move, the payoff is real but small: term (ii) is +0.20 to +1.84 points at $\alpha = 0.10$ and +0.00 to +0.65 at $\alpha = 0.15$, and it is not distinguishable from zero on two cells (POPE-adv LLaVA at $\alpha = 0.15$, $p = 0.41$; AMBER at $\alpha = 0.15$, $p = 0.32$). Term (iii) in the table, the mass *lost* from the repair set as λ opens and absorbs it, is small and negative everywhere, which is the nesting the method assumes showing up in the data.

The honest headline is therefore not leverage but cheapness: a small mass of near-certain repairs can be added to the emitted set at essentially no risk cost, and *on top of that* dilution contributes a fraction of a point to 1.8 points on the minority of splits where the constraint actually loosens. Figure 3B plots the two terms. Note that the AMBER row of Table 6 is positive: conditional on certifying, AMBER behaves like the other settings, which is the first sign that its reported loss is about something else entirely (§6).

4.8 When it helps: a checkable precondition

CCRC is not a free improvement. Its upside is bounded by what filtering leaves on the table, while its downside — perturbing a nearly-saturated constraint — is not symmetrically bounded.

Proposition 4 (Gain is capped by abstained mass)**.** *Let $\mathcal{C}_{\text{filt}}$ be the coverage certified by filtering at level (α, δ) and $\mathcal{C}_{\text{CCRC}}$ that certified by CCRC. Then*

$$\mathcal{C}_{\text{CCRC}} - \mathcal{C}_{\text{filt}} \;\leq\; 1 - \mathcal{C}_{\text{filt}},$$

Table 6: Decomposition of the CCRC gain at $\alpha = 0.10$ into the part that is automatic and the part attributable to dilution (400 paired splits, conditioned on both arms certifying; abort rates are reported separately in Table 9). Term (i) is the repaired mass available at filtering's own threshold, which is emitted whether or not the threshold moves; term (ii) is the extra accepted mass admitted because it did. "$\hat{\lambda}$ moves" is the fraction of splits on which the two arms select different thresholds. p-values are two-sided paired t-tests on term (ii).

Setting	$\hat{\lambda}$ moves	gain	(i) repairs emitted	(ii) dilution [95% CI]	p(ii)
POPE-1500 LLaVA	28.0%	+2.68	+1.82	+0.86 [+0.63, +1.09]	2×10^{-12}
POPE-adv LLaVA	29.4%	+3.59	+1.74	+1.84 [+1.22, +2.46]	1×10^{-8}
POPE-adv Qwen2-VL	19.9%	+2.03	+1.02	+1.01 [+0.43, +1.59]	7×10^{-4}
POPE-adv LLaVA+VCD	12.3%	+2.05	+0.97	+1.08 [+0.62, +1.53]	4×10^{-6}
AMBER(d) LLaVA	15.2%	+2.17	+1.97	+0.20 [+0.08, +0.31]	9×10^{-4}
term (iii), repair mass absorbed as λ opens: -0.02, -0.05, -0.03, -0.17, -0.17 points respectively					

and $1-\mathcal{C}_{\text{filt}} \geq (\mu-\alpha)/(1-\alpha)$ *by Proposition 1. No matching lower bound on* $\mathcal{C}_{\text{CCRC}}$ *holds, so the upside is bounded by the abstained mass while the downside is not.*

Proof. Coverage is at most one, so the gain cannot exceed the abstained fraction $1-\mathcal{C}_{\text{filt}}$; the floor inequality is Proposition 1 applied to the filtering policy. The asymmetry claim follows because Proposition 3 shows the emitted risk strictly increases when repairs are admitted with $r_r > r_a$, and Remark 4 shows this can move $\hat{\lambda}$ downward without bound. □

Remark 5 (Why this is a diagnostic and not a prediction). An upper bound on the gain cannot predict a loss. Write $1-\mathcal{C}_{\text{filt}} = (\mu-\alpha)/(1-\alpha)+\varepsilon$. One might hope ε is small so that headroom $\mu-\alpha$ alone governs the achievable gain, but our own data refutes it: on AMBER the floor is 1.6% while $1-\mathcal{C}_{\text{filt}} = 8.2\%$, so $\varepsilon = 6.6\%$, four times the floor. The usable statement is therefore the crude one: the gain cannot exceed the abstained mass, so a setting that already certifies most of its inputs has little to win and the same exposure to loss.

The headroom diagnostic, demoted. It is tempting to go further and read Proposition 4 as *predicting* our one negative result, with AMBER — $\mu = 11.4\%$ against $\alpha = 0.10$, floor 1.6%, filtering already certifying 91.8% — as the confirming instance. That reading does not survive §6, which shows AMBER's loss is an abort phenomenon and that conditional on certifying anything CCRC *wins* there. We therefore separate the proved bound from the unproved diagnostic and label the latter honestly:

Conjecture 1 (Headroom precondition; *no confirming instance*)**.** *There is a threshold* $h > 0$ *such that CCRC's certified-coverage gain over filtering is non-negative whenever the headroom* $\mu-\alpha$ *exceeds* h *and an applicable evidence channel exists, and that settings with* $\mu-\alpha < h$ *are at material risk of a loss.*

We state this as a conjecture because we have no evidence for it. The bound of Proposition 4 is true but is an upper bound, and an upper bound cannot predict a loss. Our own five settings do not support the conjecture even as a trend: the four positive points are not ordered by headroom (the largest headroom, VCD at 9.6 points, gives the second-smallest gain, and the second-smallest headroom, POPE-adversarial at 7.3 points, gives the largest); the one negative point is now explained by a mechanism that has nothing to do with headroom; and at $\alpha = 0.15$ Qwen has *negative* headroom ($\mu = 12.9\% < \alpha$) and still gains. With $n = 5$ settings we can neither fit h nor reject it. Anyone deploying CCRC should therefore use the two checks that *are* supported — is there abstained mass to recover, and is the calibration fold large enough that the first hypothesis of the fixed sequence can pass (§6) — and treat Conjecture 1 as an open question rather than as a screening rule.

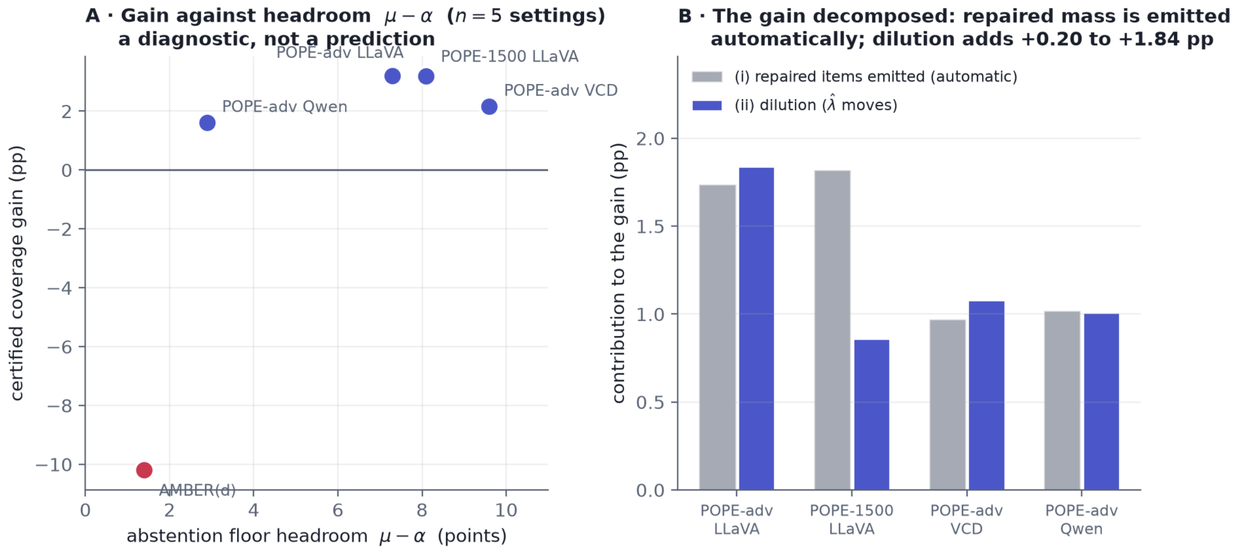

Figure 3: (A) Certified coverage gain against the abstention-floor headroom $\mu - \alpha$ at $\alpha = 0.10$, $n = 5$ settings. The panel is a diagnostic and not a prediction (Remark 5, Conjecture 1): the four positive points are not ordered by headroom, and the single negative point is explained by §6 rather than by its position on this axis. We plot it because five points is what we have, and because it is the evidence a reader would otherwise ask for. (B) The gain decomposed as in Table 6: grey bars are term (i), the repaired items, which are emitted whether or not the threshold moves; blue bars are term (ii), the dilution payoff. We do not plot gain against repaired mass with multiplier reference lines, because that ratio is automatic given term (i) and demonstrates nothing about the mechanism (§4.7). The loss cell is omitted from (B) because a decomposition of a negative gain is not informative; it is decomposed by split class in Table 15 instead.

4.9 Algorithm

5 Experiments

Protocol, and why not the tempting alternative. Every number in this paper uses one protocol: a three-way disjoint split (fit the score, calibrate the threshold, test), the fixed-sequence test of Algorithm 1 started at $k_{\min}$, Clopper–Pearson bounds at $\delta = 0.10$, and ≥ 400 *paired* random splits (the one exception is Table 14, at 60, for the reason given there) — every arm of every comparison sees byte-identical folds, so each gain is a paired difference reported with its per-split standard deviation, a 95% Student-t interval, and a two-sided paired t-test (with a two-sided Wilcoxon signed-rank test as a robustness check, which agrees everywhere). All p-values in this paper are two-sided.

Insisting on the fixed-sequence rule deserves a paragraph, because it costs coverage and because the alternative is what an implementer reaches for first: an exhaustive maximum-coverage search — "of all thresholds whose Clopper–Pearson bound passes, keep the one that emits the most". That rule has no valid theory behind it, since the selection event is a union over the whole grid and the per-threshold δ does not transfer to the selected threshold, and it is measurably more anti-conservative: on nine of ten filtering cells its population-risk exceedance is strictly higher, and it buys 0.9–13.9 points of coverage it has not paid for. We do not claim it breaks the guarantee empirically — on the cells we ran, both rules keep exceedance below δ — but it conceals the abort behaviour that turns out to be the dominant effect in three separate places (§6), because it is free to skip past the small-k thresholds where the bound tolerates no errors at all. The cost of validity is real: paired against the max-coverage rule, the fixed sequence certifies 2.2 points less on POPE-1500 and 13.9 less on POPE-adversarial at $\alpha = 0.10$. Every number in this paper is computed under the fixed-sequence rule at ≥ 400 repetitions, where the Monte-Carlo standard error on these gains is a few tenths of a point rather than the 0.8–4.2 points a 20–60-repetition estimate carries.

Algorithm 1 CCRC (calibration and deployment). Thresholds are score *values*, fixed before calibration, which is what Theorem 1 covers; the bracketed lines are the rank-indexed substitution our implementation makes (Remark 3).

Require: folds $\mathcal{D}_{\text{fit}}, \mathcal{D}_{\text{cal}}$ (disjoint), levels α, δ, repair gate value g, descending grid $t_1 > \cdots > t_M$ of score values, with t_1 chosen so that $K_{t_1} \geq k_{\min}$ in expectation (Lemma 1)

fit channel-1 score s on $\mathcal{D}_{\text{fit}}$ ▷ disjoint from calibration; see §8
[implemented variant: set $g \leftarrow Q_m(1-q)$ *and* $t_j \leftarrow Q_s(1-\lambda_j)$ *from* $\mathcal{D}_{\text{cal}}$*, with* $q = 0.10$ *and* $\lambda_1 = \min\{0.9, \max\{0.05, 1.5k_{\min}/n_{\text{cal}}\}\}$*, the factor* 1.5 *giving headroom against fold-to-fold variation in K]*
$\hat{t} \leftarrow \varnothing$
for $j = 1$ to M **do**
    $A \leftarrow \{i \in \mathcal{D}_{\text{cal}} : s_i \geq t_j\}$; $R \leftarrow \{i \notin A : m_i \geq g\}$
    $V \leftarrow \sum_{i \in A}(1 - Y_i) + \sum_{i \in R}(1 - Y_i^{(2)})$; $K \leftarrow |A| + |R|$
    if $K < k_{\min}$ **then break** ▷ untestable; skipping it would break fixed-sequence FWER control, so the grid must *start* above $k_{\min}$
    else if $U(V, K; \delta) \leq \alpha$ **then** $\hat{t} \leftarrow t_j$
    else break ▷ fixed-sequence testing stops at the first non-rejection
    end if
end for
at test time, for input x**: if** $\hat{t} = \varnothing$ abstain on every input (coverage 0); **else if** $s(x) \geq \hat{t}$ emit $\hat{a}(x)$; **else if** $m(x) \geq g$ emit $b(x)$; **else** abstain
▷ $\hat{t}$ and g are fixed before deployment and never recomputed at test time; in the implemented variant they are the calibration-fold quantiles frozen above

How validity is audited, and what the p-values mean. Mean realised risk is *not* the guarantee. The claim is $\Pr[\mathcal{R} \leq \alpha] \geq 1 - \delta$, so the quantity to audit is the fraction of splits whose realised risk exceeds α, compared against δ, and mean risk appears nowhere in this paper. We report two versions, because realised risk on a finite test fold carries binomial noise that is not a validity failure. Scored on the held-out test fold ($n/3$ items), exceedance is 3.1–22.3% across the 40 cells of Table 20 against $\delta = 0.10$; scored on the full item set — a lower-noise proxy for the population risk of the selected policy, and the column that decides the question — it is 0.0–5.1%. The guarantee holds; the discrepancy is finite-fold noise around a population risk that Clopper–Pearson deliberately parks just under α, and it appears for the plain filtering baseline as much as for any repair arm. Aborted splits emit nothing, so they have no risk and are excluded from these denominators; the abort rate is printed beside every exceedance figure.

One caveat on every interval and p-value here, since it would otherwise be over-read. The 400 splits resample the *same* items, so the interval shrinks like $1/\sqrt{\text{reps}}$ while dataset-level uncertainty does not shrink at all. These statistics test whether the mean gain over the split distribution differs from zero *conditional on this dataset*. They are the right statistic for "is arm A better than arm B on these items" and the wrong one for "would this replicate on a fresh benchmark"; the per-split standard deviation, which we also report, is the honest measure of spread.

Models and data. Backbones are LLaVA-1.5-7B (Liu et al., 2024c) in 4-bit and Qwen2-VL-2B-Instruct (Wang et al., 2024); we additionally evaluate LLaVA decoded with visual contrastive decoding (Leng et al., 2024). Channel 2 is OWLv2-base (Minderer et al., 2023). Benchmarks are POPE (Li et al., 2023) (random and adversarial splits), AMBER discriminative (Wang et al., 2023a), MME (Fu et al., 2023), GQA (Hudson & Manning, 2019) and HallusionBench (Guan et al., 2024). All per-item extracted signals are released so that the analyses can be reproduced without a GPU.

5.1 The correctness score

Validity holds for any score; only efficiency depends on which one is used. We therefore first characterise what makes a score efficient here.

Table 7: Selective prediction on POPE (LLaVA-1.5-7B, $n = 1500$, 400 paired splits). AURC is area under the risk–coverage curve (lower better). "abort" is the fraction of splits on which the fixed sequence certifies nothing and the deployed system covers 0%; coverage is the mean over all splits including those, so a large abort rate is visible as a large standard deviation. Gains in the last column are paired differences against the confidence-only row with 95% CIs; all have $p < 10^{-40}$ (two-sided paired t). The validity audit for these arms is Table 20.

Score	AURC ↓	cov@10% ↑	abort	gain over row 2
Raw confidence	0.0774	$20.2 \pm 27.6\%$	59.2%	—
Learned, confidence only	0.0682	$41.6 \pm 29.5\%$	32.5%	—
+ CLIP grounding (Radford et al., 2021)	0.0637	$63.4 \pm 7.4\%$	0.5%	+21.8 [+18.9, +24.6]
+ detection grounding (Minderer et al., 2023)	0.0554	$68.6 \pm 9.0\%$	0.8%	+27.0 [+24.2, +29.8]
+ both	**0.0547**	$\mathbf{70.7 \pm 5.6\%}$	**0.0%**	+29.1 [+26.2, +31.9]

Table 8: Against a faithful self-consistency baseline on POPE-adversarial, the hardest split ($K = 3$ samples, LLaVA-1.5, $n = 450$, 400 paired splits, $\alpha = 0.10$). This arm is *not* ConfLVLM: ConfLVLM's scorer is CLIP image–text similarity, which appears as the "+ CLIP grounding" row of Table 7 and as rung 1 of Table 13.

Score	AURC ↓	cov@10% ↑	abort
Self-consistency only	0.1366	$0.0 \pm 0.0\%$	100.0%
Confidence + self-consistency	0.0770	$11.0 \pm 21.1\%$	75.2%
+ detection grounding (ours)	**0.0587**	$\mathbf{34.8 \pm 27.5\%}$	**32.2%**

Table 7 compares scores on POPE. An open-vocabulary *detection* signal is worth +27.0 points of certified coverage over a learned combination of the model's own confidence signals, against +21.8 for generic image–text similarity (Radford et al., 2021), both with intervals far from zero. Magnitudes that large are not a matter of better ranking but of the abort column: under a valid fixed sequence the confidence-only baseline *certifies nothing at all* on 32.5% of splits and raw confidence on 59.2%, so what a weak score costs is not a few points of coverage but the entire deployment on a third of calibration draws. The right statement is therefore not "a better score buys a few more points" but "a weak score cannot certify at all", and §6 gives the mechanism exactly.

The AUROC column tells the same story more weakly and is omitted for space: it moves from 0.768 (raw confidence) through 0.786, 0.799 and 0.832 to 0.834, i.e. the ranking is identical and the differences are small, which is the point — what the fixed sequence rewards is separability in the upper tail, not average ranking quality.

Against a baseline given *both* confidence and multi-sample self-consistency — the scoring philosophy of Li et al. (2025) — detection grounding adds +23.8 points (95% CI [+21.4, +26.3]) on the adversarial split (Table 8). The interval excludes zero comfortably, but the honest reading is again about aborts and not about coverage: the baseline fails to certify anything on 75.2% of splits and the self-consistency-only score on 100%, because $n = 450$ leaves a 150-item calibration fold against $k_{\min} = 22$. The right sentence is *the baseline cannot certify on this split at all*, not *grounding adds* 24 *points to a working baseline.*

Two regularities follow, one of which needs a caveat. The advantage of grounding *grows as the guarantee tightens*: at $\alpha = 0.05$ certified coverage goes 9.0% → 27.3%, a three-fold increase, with the abort rate falling 65.8% → 24.0%, while at $\alpha = 0.20$ both scores certify almost everything (97.3% vs. 97.7%; Figure 4). The caveat is that "the grounded score dominates at every level" holds for detection grounding but not for CLIP, whose $\alpha = 0.20$ gain is −0.06 points ($p = 0.27$), indistinguishable from the baseline.

The second regularity is that the advantage tracks how much the base model hallucinates, and that it can reverse sign. On the more accurate Qwen2-VL-2B the same addition is worth +3.10 points [+1.91, +4.30] at $\alpha = 0.10$ rather than +27.0 — but at $\alpha = 0.15$ it *costs* −0.50 points [−0.85, −0.14], $p = 0.006$. Neither arm aborts on any split in that cell, so the negative result cannot be explained away as an abort artefact;

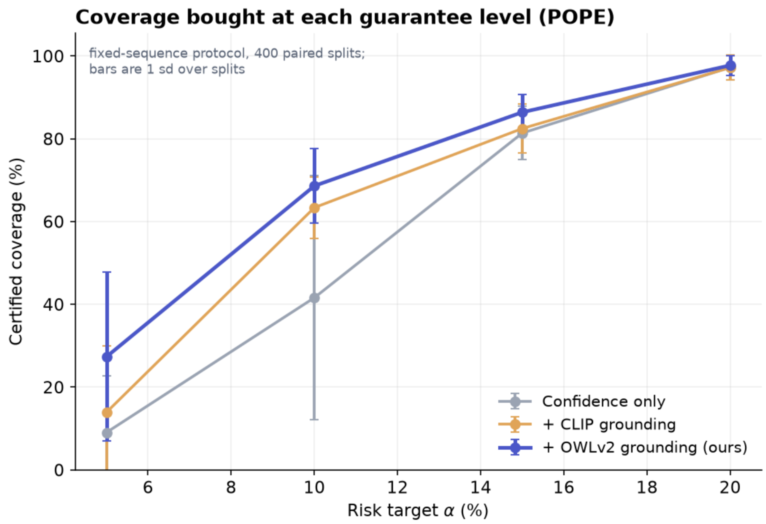

Figure 4: Certified coverage as a function of the risk target (POPE-1500, 400 paired splits). The detection-grounded score dominates at every level and the margin widens as α tightens; at $\alpha = 0.05$ it roughly triples certified coverage. Error bars are one standard deviation over splits, and they are large at $\alpha = 0.05$ because the distribution is bimodal — a large fraction of splits certify nothing at all (abort rates 65.8%, 49.8% and 24.0% for the three arms at $\alpha = 0.05$). That bimodality is the finding, not noise to be smoothed away.

grounding genuinely does not pay on an already-accurate backbone at a loose target. Grounding is most valuable exactly where selective prediction is most expensive, and is not free where it is cheap.

Remark 6 (accuracy is not separability). A detector whose standalone accuracy matches the VLM's on POPE (83.1% vs. 81.9%) certifies far less coverage when used as a selective predictor in its own right (60.1% vs. 68.6%). Conformal coverage depends on how well a score *separates* correct from incorrect outputs, not on how often the underlying predictor is right. This also answers the natural objection to CCRC — "if channel 2 is good, why not just use it?" — in all five settings (Table 12).

5.2 CCRC across settings

Table 9 reports CCRC against filtering under an identical protocol, with the paired interval, the two-sided paired test, and the abort rate for every cell. Certified coverage rises by +0.40 to +7.59 points in eight of ten cells, and *every* one of those intervals excludes zero; it falls by 10.20 and 11.90 points on the two AMBER cells, whose cause is §6 rather than the headroom precondition. The guarantee holds in every row (Table 20).

Two explanations of the AMBER loss that we can rule out. It is not a weak repair channel: the detector is 95.6% accurate on AMBER overall and 100% accurate in the top-decile region repairs are drawn from, and the repair region's realised risk there is 3.6% at $\alpha = 0.10$ and 0.0% at $\alpha = 0.15$ — *below* the accepted region's 4.5% and 7.5%, so by Proposition 3 dilution is running in CCRC's favour on the setting where it loses.

Nor is it absent headroom. The obvious way to test that is to subsample POPE to AMBER's $n = 228$ and ask whether the loss follows the sample size rather than the headroom, and it matters which contrast is subsampled. The *grounding-score* gain (confidence-only against confidence-plus-detection) is a different quantity and cannot bear on the question at all; for the record it is +0.2 points at $n = 228$, switching on only at $n \geq 700$ (+2.9 at $n = 400$, +19.6 at 700, +26.4 at 1500) once the calibration fold is large enough to certify anything. The comparison that *is* relevant — CCRC against filtering on POPE subsampled to $n = 228$ — *loses* 3.65 points, with abort rates of 43.5% and 53.8%. Sample size is the explanation, and §6 gives it properly.

Table 9: CCRC (q=0.10) versus filtering, 400 paired splits per cell. "ab" is the abort rate: the fraction of splits on which that arm certifies nothing and covers 0%. Gain is the mean paired difference with a 95% Student-t interval; p is a two-sided paired t-test on the same 400 differences (a two-sided Wilcoxon signed-rank test agrees in every row). The validity audit — Pr[realised risk $> \alpha$] against $\delta = 0.10$, which is the guarantee, rather than mean risk, which is not — is in Table 20. Coverage means include aborted splits as 0%, which is what a deployer experiences.

Dataset	Backbone	n	μ	α	Filter (ab)	CCRC (ab)	Gain [95% CI]	p
POPE	LLaVA-1.5	1500	18.1%	0.10	68.6% (0.8)	**71.8%** (0.0)	+3.18 [+2.6, +3.8]	8×10^{-22}
POPE-adv	LLaVA-1.5	444	17.3%	0.10	44.0% (18.8)	**47.2%** (17.5)	+3.19 [+1.7, +4.7]	5×10^{-5}
POPE-adv	Qwen2-VL-2B	591	12.9%	0.10	77.9% (0.2)	**79.5%** (0.8)	+1.60 [+0.8, +2.4]	2×10^{-4}
POPE-adv	LLaVA+VCD	591	19.6%	0.10	48.8% (5.8)	**51.0%** (4.0)	+2.15 [+1.3, +3.0]	9×10^{-7}
POPE	LLaVA-1.5	1500	18.1%	0.15	86.4% (0.0)	**88.4%** (0.0)	+1.95 [+1.8, +2.1]	6×10^{-102}
POPE-adv	LLaVA-1.5	444	17.3%	0.15	69.1% (9.5)	**76.7%** (0.5)	+7.59 [+5.5, +9.7]	5×10^{-12}
POPE-adv	Qwen2-VL-2B	591	12.9%	0.15	94.8% (0.0)	**95.2%** (0.0)	+0.40 [+0.3, +0.5]	1×10^{-26}
POPE-adv	LLaVA+VCD	591	19.6%	0.15	75.8% (0.8)	**77.2%** (0.2)	+1.39 [+0.5, +2.3]	0.0022
AMBER(d)	LLaVA-1.5	228	11.4%	0.10	**91.8%** (0.0)	81.6% (**13.0**)	−10.20 [−13.4, −7.0]	7×10^{-10}
AMBER(d)	LLaVA-1.5	228	11.4%	0.15	**95.9%** (0.0)	84.0% (**12.8**)	−11.90 [−15.1, −8.7]	1×10^{-12}

The POPE-adv rows use the 444 (LLaVA) and 591 (Qwen, VCD) items for which the detector returns a value and the four canonical features. Table 11's POPE-adv rows are a *different* quantity — all 450/600 items with ungrounded detector values imputed to the median, plus a self-consistency feature — and the two should not be reconciled.

5.3 A polarity-symmetric repair gate: valid, precise, and useless

Remark 1 records the sharpest scope restriction on this paper: our margin $m = |o - \theta|$ gives the two polarities unequal dynamic range, so every admitted repair says "the object is present" and CCRC as evaluated cannot touch the false-positive existence hallucination that POPE-adversarial exists to elicit. The obvious remedy is a polarity-symmetric gate. We evaluate it here, and the result is worth a subsection because it converts an admitted gap into a measured impossibility.

Construction, and why it costs nothing in δ. Normalise each polarity onto $[0, 1]$ separately, $m^+ = (o - \theta)/(1 - \theta)$ for detector positives and $m^- = (\theta - o)/\theta$ for negatives, and pre-specify *two* gates at the *same* fixed $q = 0.10$, one per polarity: $g^+ = Q_{1-q}(m^+ \mid o \geq \theta)$ and $g^- = Q_{1-q}(m^- \mid o < \theta)$, with repair when the item clears the gate for its own polarity. Because the repair mask depends only on the calibration fold and on o — never on the acceptance threshold — the emitted family is still $\mathcal{E}_t = A_t \cup M$ with M constant in t, so it is still nested, a single fixed-sequence test at *full* δ is still valid, and no union bound or $\delta/2$ split is introduced. We verified this numerically over a λ-grid on 30 random splits per setting: zero nesting violations in every arm. The construction also does not enlarge the repair budget, it redistributes it: admitting the top q of each polarity keeps total gate mass at roughly q of items.

A single gate on the pooled two-sided normalised margin — the other obvious construction — is the wrong one, and instructively so: the negative-side margin distribution is concentrated near 1 (its 90th percentile is 0.947–0.973 across settings) while the positive side's is not (0.555–0.582), so a global top-q fills with negatives and the repaired mass becomes 100% absent-polarity in every cell. It inverts the bias rather than removing it. We report the two-gate version.

It admits negative repairs, at near-perfect precision. It works as designed on its own terms (Table 10). Negative-polarity repairs are admitted — 0.05–0.33% of test items, against exactly 0.00% for the one-directional gate in all ten cells — and their precision is 99.9–100% wherever any are admitted. Certified coverage improves over the one-directional gate on the three POPE-adversarial cells at $\alpha = 0.10$ (+1.10 to +4.61 points, $p = 0.040$, 0.0024, 1.2×10^{-6}), and a decomposition against a positive-branch-only arm confirms that on those cells the improvement *is* the new negative repairs (+1.87 to +6.81 points, all $p < 10^{-5}$). It does nothing at $\alpha = 0.15$ on POPE (three of four differences not significant) and it costs 1.1–1.3 points on POPE-1500 at both risk levels.

And it repairs no false positives at all. The reason the symmetric gate was wanted was to correct the canonical false-positive existence hallucination: the VLM says "yes", the object is absent, and the detector says "no". Counting exactly those items among the repaired ones gives 0.000% in every setting at both risk levels, for both the one-directional and the symmetric gate — ten cells, no exceptions. The admitted negative repairs are almost all correct and almost all land on items the model *already* answered "no" correctly. They change no answer. They are pure risk ballast, which is to say they are the dilution mechanism of §4.7 operating on the other polarity, and nothing more.

The reason is structural and can be seen without any calibration. Rank the detector-negative items by confidence and ask where the repairable false positives sit. On POPE-1500 the top decile of the negative side is 97.4% accurate and contains **zero** of the 46 reachable false positives (of 85 in total); the top half is 83.3% accurate and contains 9; the remaining 37 sit in the bottom half, whose last decile is only 36.7% accurate. The same shape holds on POPE-adv LLaVA (top decile 100% accurate, 0 of 29 reachable), Qwen2-VL (0 of 20 in the top six deciles) and VCD (0 of 78 in the top two deciles). **On the negative side, repair opportunity and repair precision are anti-correlated.** An object scoring $o \approx 0.001$ is so obviously absent that the VLM also says "no"; the VLM only hallucinates presence when there is *some* visual evidence, which is exactly where the detector's own margin is smallest and its accuracy falls to 35–55% — far below $1 - \alpha$, so those items can never be certified. No strict negative gate can reach the hallucinations, and any gate loose enough to reach them is uncertifiable. This is a property of the signal, not of the gate's parameterisation, which is why we state it as a limitation of certified correction with a detector channel rather than as an implementation detail. The one exception is AMBER, where the detector is accurate all the way down the negative side and a loose negative gate could reach 10–12 of the 14 false positives at full precision — and AMBER is the one setting where CCRC should not be used at all.

An independent confirmation of the AMBER diagnosis. The symmetric gate's largest apparent effect is on AMBER, where it converts the one-directional gate's loss into gains of +0.71 and +0.32 points over filtering on the same 500 splits. It would be easy to sell that as the symmetric gate rescuing the paper's negative result. It is not. Decomposing on identical splits, the AMBER improvement is +10.75 and +12.28 points from *re-scaling the positive gate* (the two-gate construction makes the positive gate much stricter, cutting repaired mass from 1.69% to 0.51%) and +0.01 and +0.00 points from polarity. What the stricter gate does is stop perturbing AMBER's smallest, most fragile first hypothesis — the abort rate falls from 13.0% to 0.0%. That is the same mechanism §6 identifies, arrived at from a different direction, and we report it as confirmation rather than as a rescue.

Verdict. The symmetric gate is valid, is free in δ, admits negative-polarity repairs at 99.9–100% precision, and helps on three of ten cells while hurting on two. It does not do the thing it was wanted for, in any setting, at either risk level. We would rather report that than leave the remedy standing as an unevaluated promise, and we regard the anti-correlation between negative-side opportunity and negative-side precision as the more useful finding: it tells anyone building certified correction on a detector channel which half of the hallucination taxonomy is out of reach and why.

5.4 Where grounding does not apply

Benchmark composition determines whether channel 2 exists at all. Table 11 reports the full sweep. On MME's full split, GQA's yes/no subset and HallusionBench, the questions are largely about attributes, counting or reasoning, so no object can be extracted for the detector to check; the procedure reduces to filtering, which remains valid. HallusionBench is instructive: LLaVA scores 51.2%, essentially chance, and the certified coverage is 0.0%. That is the guarantee working as intended — with $\mu \approx 0.49$ and $\alpha = 0.10$, Proposition 1 forces abstention on at least 43% of inputs, and in practice on all of them.

5.5 Composition with a mitigation decoder

Because the two programmes of §2 are orthogonal — mitigation lowers μ, risk control governs what is emitted — our layer should compose with a mitigation decoder. It does, and most strongly where the decoder's own confidence is poorly calibrated. On POPE-adversarial with visual contrastive decoding (Leng et al., 2024),

Table 10: Polarity-symmetric repair gate against the one-directional gate (500 paired splits; all arms share splits and the fitted score; q fixed at 0.10 throughout, so no confidence budget is spent on the choice). "rep. mass" is the fraction of the test fold repaired, split by polarity. "FP corrected" is the fraction of test items that were false-positive existence hallucinations *and* were corrected. The validity audit gives $\Pr[\text{risk} > \alpha] \leq 0.032$ on the population proxy for every arm in every cell. *One caveat on cross-table comparison*: this experiment draws its own 500 paired splits so that the symmetric and one-directional arms can be contrasted pairwise, so its one-directional column differs from Table 9's CCRC column in the first decimal (e.g. 46.76 here against 47.2 there for POPE-adv LLaVA at $\alpha = 0.10$, and 81.49 against 81.6 on AMBER). Only the within-table differences are paired and only they should be read as effects; the levels agree with Table 9 to Monte-Carlo error.

		certified coverage		rep. mass (pos / neg)		FP corrected	
Setting	α	one-dir	sym-2gate	one-dir	sym-2gate	one-dir	sym
POPE-1500 LLaVA	0.10	**71.71**	70.60	1.80 / 0.00	0.78 / 0.10	0.000%	0.000%
POPE-1500 LLaVA	0.15	**88.48**	87.23	1.33 / 0.00	0.48 / 0.00	0.000%	0.000%
POPE-adv LLaVA	0.10	46.76	**51.37**	1.51 / 0.00	0.53 / 0.33	0.000%	0.000%
POPE-adv LLaVA	0.15	76.87	**77.25**	1.33 / 0.00	0.36 / 0.05	0.000%	0.000%
POPE-adv Qwen2-VL	0.10	79.10	**80.66**	0.96 / 0.00	0.42 / 0.14	0.000%	0.000%
POPE-adv Qwen2-VL	0.15	**95.09**	95.08	0.35 / 0.00	0.15 / 0.00	0.000%	0.000%
POPE-adv LLaVA+VCD	0.10	50.31	**51.42**	0.86 / 0.00	0.22 / 0.30	0.000%	0.000%
POPE-adv LLaVA+VCD	0.15	**76.72**	76.47	0.43 / 0.00	0.17 / 0.05	0.000%	0.000%
AMBER(d) LLaVA	0.10	81.49	**92.25**	1.69 / 0.00	0.51 / 0.01	0.000%	0.000%
AMBER(d) LLaVA	0.15	83.90	**96.18**	0.39 / 0.00	0.31 / 0.00	0.000%	0.000%

Table 11: Full dataset sweep at $\alpha = 0.10$, 400 paired splits. "grnd" indicates whether channel 2 applies. Coverage columns are confidence-only $\rightarrow$ with grounding, each with its abort rate in brackets; the last column is the paired grounding gain with a 95% CI. The POPE-adv and AMBER rows use all items with ungrounded detector values imputed to the median plus a self-consistency feature, and are therefore not comparable with Table 9's grounded-item rows.

Dataset	Backbone	acc	grnd	cov@10% (abort)	grounding gain
POPE (1500)	LLaVA-1.5	81.9%	yes	41.6 (32.5) → **68.6** (0.8)%	+27.0 [+24.2, +29.8]
POPE-adv (450)	LLaVA-1.5	82.7%	yes	11.7 (75.0) → **36.5** (29.8)%	+24.9 [+22.2, +27.6]
POPE-adv (600)	Qwen2-VL-2B	87.3%	yes	75.9 (0.0) → **79.0** (0.0)%	+3.1 [+1.9, +4.3]
POPE-adv (600)	LLaVA+VCD	80.3%	yes	1.7 (95.8) → **47.3** (5.0)%	+45.6 [+43.8, +47.4]
MME-existence[a]	LLaVA-1.5	95.0%	yes	0.0 (100) → 0.0 (100)%	±0.0
AMBER(d)	LLaVA-1.5	78.6%	partial	see Table 9	
MME (full, 700)	LLaVA-1.5	68.9%	no	2.6 (85.0)%	—
GQA (yes/no, 700)	LLaVA-1.5	72.4%	no	4.5 (84.0)%	—
HallusionBench (447)	LLaVA-1.5	51.2%	no	0.0 (100)%	—

[a] Only 60 items, leaving a 20-item calibration fold under the three-way split, which is below $k_{\min} = 22$ (Lemma 1), so both arms abort on every split. Calling such a cell "underpowered" understates it; "100% abort" is the same fact stated so that a reader can see it. Sample size is not the only obstruction: at this subset's low base error rate $U(3, 60; \delta) = 0.108 > \alpha$, so power would bind even with the full 60 items available for calibration.

adding grounding improves AURC from 0.1031 to 0.0630 and certified coverage from 1.7% to 47.3%, a paired gain of +45.6 points [+43.8, +47.4] — the largest score effect anywhere in this paper, and almost entirely an abort effect: the confidence-only arm certifies nothing on 95.8% of splits against 5.0% for the grounded arm (Table 11). VCD improves the decoder's answers while leaving its logits a poor correctness score, which is exactly the situation in which an external channel is worth the most.

Table 12: Detector-only selective prediction versus CCRC ($\alpha = 0.10$, identical protocol, 400 paired splits). "Detector only" emits the detector's answer, gated on its own evidence margin.

Setting	VLM acc	det. acc	Filter (VLM)	CCRC	Detector only
POPE-1500 LLaVA	81.9%	83.1%	68.6%	**71.8%**	60.1%
POPE-adv LLaVA	82.7%	82.4%	44.0%	**47.2%**	17.7%
POPE-adv Qwen2-VL	87.1%	82.9%	77.9%	**79.5%**	22.9%
POPE-adv LLaVA+VCD	80.4%	82.9%	48.8%	**51.0%**	22.9%
AMBER(d) LLaVA	88.6%	95.6%	**91.8%**	81.6%	19.2%[†]

[†] The detector is 95.6% accurate on AMBER and would certify nearly everything given enough calibration data, but with $n = 228$ a three-way split leaves 76 calibration items and the fixed sequence largely aborts (a single error at the grid start already breaks the Clopper–Pearson bound at $k \approx 33$; §6). The figure therefore measures calibration-fold size, not detector quality; we report it rather than a large-calibration number because it is what this protocol yields. An earlier draft of this paper reported 92.6% here and concluded that a detector-only baseline beats CCRC on AMBER; that number is not reproducible under any split and the claim is withdrawn.

5.6 Baselines and attribution

Detector alone. Table 12 answers Remark 6's objection directly. CCRC beats a detector-only selective predictor in all five settings, mostly by very large margins, because the detector's correctness is harder to *score* than the VLM's: its evidence margin is a one-dimensional signal that saturates, so it separates its own correct from its own incorrect answers poorly even where its accuracy is high. On AMBER the detector is more accurate than the VLM (95.6% vs. 88.6%) and still certifies only 19.2%.

A published scorer, and where our margin comes from. Table 13 places the CLIP-similarity scoring function of Li et al. (2025) in our pipeline and adds our components one at a time, cumulatively, so that each row is the previous row plus one thing and the increments sum to the total. The aggregate margin is +66.8 points; +64.5 of it is the correctness score and +2.3 is certified correction. Reporting the aggregate alone would misattribute the paper, so we report the ladder, and the number we ask a reader to hold us to is the +2.3: it is small, it is the only part that is ours as an algorithm, and it is measured under a protocol that does not flatter it. We also note the comparison understates Li et al. (2025), whose method was designed for free-form caption claims rather than yes/no probes; we transplant their scorer, not their system. Figure 5 shows the corresponding risk–coverage curves.

5.7 Composition with the closest concurrent method

Xu et al. (2026) rescue borderline claims by acquiring zoomed views and re-scoring the *original* assertion. Holding the base score fixed and adding one mechanism per arm (Table 14), the two mechanisms are complementary: composing them beats every single-mechanism arm at both risk levels. Re-reading the image and replacing the answer rescue different claims. We do not claim to beat Xu et al. (2026): our reproduction uses three coarse crops and under-delivers relative to their published gain at $\alpha = 0.10$, so the defensible conclusion is composition, not superiority.

6 A small-calibration failure mode of fixed-sequence testing

Three results in this paper looked unrelated until we measured the abort rate, and are in fact one mechanism seen three times. Because it is elementary, checkable in advance, and — on our data — larger than the effect our method is designed to produce, we give it its own section rather than three scattered paragraphs.

6.1 The mechanism

Two facts about the procedure combine badly at small calibration size. The first is Lemma 1: with $\alpha = \delta = 0.10$ no emitted set smaller than $k_{\min} = 22$ items can be certified at all, and the value is *bound-dependent* — 22 for Clopper–Pearson but 44, 71 and 116 for the betting, empirical-Bernstein and Hoeffding bounds

Table 13: Attribution on POPE-1500 ($\alpha = 0.10$, 400 paired splits). The ladder is cumulative: each row adds one component to the row above, and the Δ column sums to the total. Most of the margin over a published scorer comes from the score, not from our procedure. Rungs 2–4 coincide exactly with rows 2, 3 and 5 of Table 7, which they must, being the same scores on the same items under the same rule. Rung 5 applies CCRC on top of the six-feature score of rung 4 and so reads 73.0%; Table 9's CCRC column reads 71.8% because it uses the four-feature canonical score, which is the configuration we report as the method. The +2.3-point increment is measured within this ladder, and the corresponding four-feature increment is +3.18 points (Table 9); we quote the smaller figure in the attribution because it is the one measured against the published scorer's own comparison.

Method	Coverage	Δ
1. CLIP-similarity scorer (Li et al., 2025), filtering	6.2%	—
2. + VLM confidence (replacing CLIP)	41.6%	+35.4
3. + CLIP grounding	63.4%	+21.8
4. + detection grounding	70.7%	+7.3
5. + certified correction (CCRC)	**73.0%**	+2.3
total margin over rung 1		**+66.8**
of which the correctness score (rungs 2–4)		+64.5
of which certified correction (rung 5)		+2.3

Table 14: One common base score, one mechanism added per arm (POPE, LLaVA-1.5, $n = 394$ grounded items, α as shown). All arms valid. Composition beats every single mechanism. This is the one table in the paper at 60 rather than ≥ 400 paired splits, because the arms require a re-scoring pass per split; it uses the same fixed-sequence rule as everything else, and the differences here are large enough that the repetition count is not the binding uncertainty.

Arm	cov@0.10	cov@0.15
Base filter (confidence only)	5.5%	16.5%
+ evidence acquisition (Xu et al., 2026)	5.5%	24.1%
+ detection-grounded score (ours)	32.1%	63.2%
+ certified correction (ours)	34.5%	71.8%
Composed	**35.8%**	**74.2%**

(Table 18) — so a grid start tuned for one bound silently disables another. The second is that fixed-sequence testing must stop at the *first* non-rejection (§4.3), and the first hypothesis in an ascending sweep is the one with the smallest emitted set.

Put together: the strictest hypothesis is simultaneously the one with the least statistical power and the one whose failure costs everything. The arithmetic is stark. Our grid starts at $\lambda_1 = 1.5k_{\min}/n_{\text{cal}}$, so the first hypothesis emits $k \approx 33$ items, and Clopper–Pearson at $k = 33$ tolerates exactly **zero** errors:

$$U(0, 33; 0.10) = 0.0674 \leq \alpha, \qquad U(1, 33; 0.10) = 0.1128 > \alpha, \qquad U(2, 33; 0.10) = 0.1533 > \alpha.$$

So whenever the top 33 calibration items by score contain a single error, the whole sequence returns nothing and the deployed system covers 0%. Whether that happens is a property of the score's extreme upper tail on a n_{cal}-item fold — not of its average quality, its AUROC, or the dataset's difficulty.

The condition for this to be fatal rather than merely costly is exactly the one Remark 4 anticipates: feasibility must be a *prefix* of the grid for "stop at the first failure" to lose nothing, and that is risk monotonicity in λ, which §4.3 declines to assume. Where feasibility is not a prefix, the first hypothesis decides the outcome by itself. On POPE-1500 with raw confidence we can verify this exactly: the measured first-hypothesis failure rate is 59.2% and the measured abort rate is 59.2% — identical, to the split.

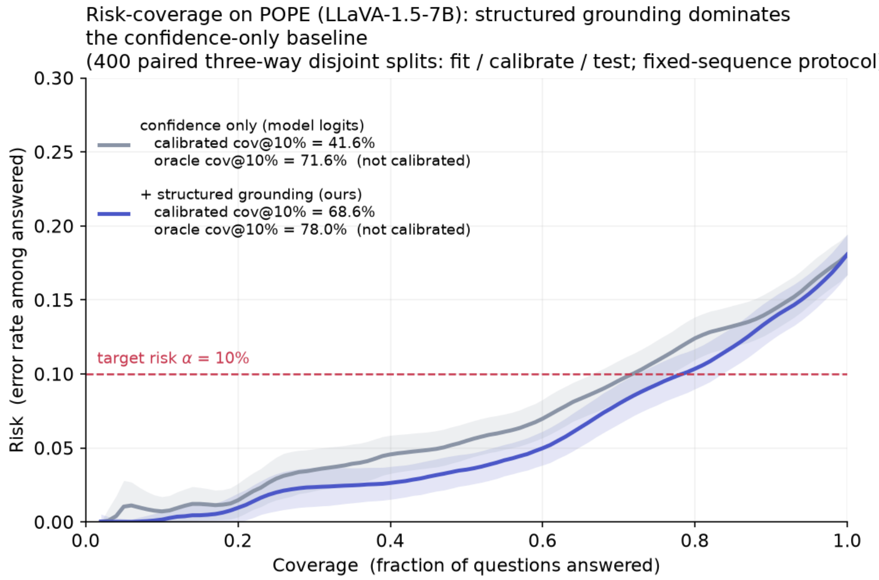

Figure 5: Risk–coverage curves on POPE-1500. The grounded score (blue) lies below the confidence-only baseline at 99% of coverage levels, meeting it only at full coverage where both necessarily equal the base risk: lower error for the same coverage, and more coverage for the same error target. Shaded bands are one standard deviation over 400 paired three-way splits; the dashed line marks $\alpha = 0.10$. The legend reports two quantities that should not be confused: *oracle* coverage read off the test-fold curve (71.6% → 78.0%), and conformally *calibrated* coverage (41.6% → 68.6%), the latter being what the tables report and the gap between them being the price of calibrating rather than knowing. The baseline is LLaVA's own output logits with no external evidence, which is why it is labelled "confidence only (model logits)" rather than "ConfLVLM-style": ConfLVLM's scorer is CLIP image–text similarity, which appears separately as rung 1 of Table 13 and as the "+ CLIP grounding" arm of Figure 7.

The max-over-thresholds rule of §5 hides all of this, because it is free to skip to a large-k threshold where dozens of errors are tolerable. That is the multiplicity it never pays for, and the abort column is what appears when the bill is settled.

6.2 Instance 1: a weak score does not certify a little less, it certifies nothing

Under a valid fixed sequence, score quality expresses itself almost entirely through the abort rate rather than through coverage given certification. Raw confidence aborts on 59.2% of splits and a learned confidence-only score on 32.5%, against 0.8% for the same score plus detection grounding (Table 7); the self-consistency baseline of Table 8 aborts on 75.2% and self-consistency alone on 100%; the confidence-only arm on VCD-decoded LLaVA aborts on 95.8% (Table 11); and at $\alpha = 0.05$ even the grounded score aborts on 24.0% (Table 19). The claim this supports is stronger than a coverage claim: the value of an external evidence channel is not that it buys a few points of coverage but that it makes certification possible at all on between a third and all of calibration draws.

Two cells in this paper have a 100% abort rate (MME-existence, HallusionBench). Their certified coverage of 0.0% is the guarantee working exactly as designed — on HallusionBench LLaVA is at chance, $\mu \approx 0.49$, and Proposition 1 forces abstention on at least 43% of inputs — but "0.0% coverage" and "mean coverage 2.6% over splits, of which 85% are zeros" are very different statements to a deployer, and only the second is honest.

Table 15: The AMBER(d) loss, decomposed (400 paired splits, $q = 0.10$). The unconditional gain is what Table 9 reports and what a deployer experiences. The conditional gain is what CCRC does when it certifies at all. The gap is entirely one split class. POPE-1500 is shown for contrast: there, CCRC never aborts and filtering occasionally does — the mirror image.

	AMBER(d), $n = 228$		POPE-1500, $n = 1500$	
	$\alpha = 0.10$	$\alpha = 0.15$	$\alpha = 0.10$	$\alpha = 0.15$
unconditional gain (Table 9)	−10.20	−11.90	+3.18	+1.95
gain | neither arm aborts	**+2.17**	**+0.47**	+2.68	+1.95
(95% CI)	[+1.8, +2.5]	[+0.4, +0.5]	—	—
abort rate, filtering	0.00%	0.00%	0.75%	0.00%
abort rate, CCRC	**13.00%**	**12.75%**	**0.00%**	**0.00%**
$\hat{\lambda}$ moves	15.2%	1.7%	28.0%	23.5%
r_a / r_r	4.53 / **3.60**%	7.45 / **0.00**%	7.96 / 0.24%	12.65 / 0.32%
dilution direction	favourable	favourable	favourable	favourable
contribution to the -10.20 *at* $\alpha = 0.10$*, by split class:*				
both certify (348 splits)	+1.89			
CCRC aborts, filter does not (52)	**−12.09**			
filter aborts, CCRC does not (0)	+0.00			

6.3 Instance 2: self-repair is defeated by the prefix, not by the region

§4.4 is the same mechanism with a fixed-mass contaminant. The self-repair block has mass independent of λ, so at the start of the grid it is the largest fraction of the emitted set it will ever be, and its 4.9–62.4% error rate fails the first hypothesis. Coverage does not degrade gracefully; it collapses, by 49.3, 37.3, 76.9 and 6.6 points, with abort rates of 73.0%, 88.8%, 98.8% and 25.2% (Table 3). The budget in (3) would in fact be satisfiable at later λ on some of these settings; the test never gets there. Note that the ordering of the four collapse magnitudes is the ordering of the four abort rates, not the ordering of the four flip-region risks.

6.4 Instance 3: this, and not headroom, is why CCRC loses on AMBER

AMBER(d) has $n = 228$ items, so a three-way split leaves a 76-item calibration fold. With $k_{\min} = 22$ the grid starts at $\lambda_1 = 1.5 \times 22/76 = 0.434$ and the first hypothesis again emits about 33 items, tolerating zero errors. AMBER's filtering arm clears that bar reliably because its score is clean at the top: it aborts on 0.00% of splits at both risk levels. Adding a fixed-mass repair block puts the occasional repair error into that first, smallest, most fragile hypothesis — and the fixed sequence must stop there. CCRC aborts on 13.00% and 12.75% of splits.

Table 15 decomposes the reported loss by split class, and the answer is unambiguous. Conditional on both arms certifying something, CCRC *wins* on AMBER: +2.17 points at $\alpha = 0.10$ ($p = 6 \times 10^{-28}$) and +0.47 at $\alpha = 0.15$ ($p = 7 \times 10^{-30}$). The splits on which CCRC aborts and filtering does not — 52 of 400 — contribute −12.09 points, which is 119% of the entire reported loss at $\alpha = 0.10$ and 103% at $\alpha = 0.15$. Every other split class contributes positively. And the dilution direction is favourable throughout: $r_r = 3.60\% < r_a = 4.53\%$ at $\alpha = 0.10$ and $0.00\% < 7.45\%$ at $\alpha = 0.15$, so the "repairs consume slack" branch of Proposition 3 — the branch a headroom explanation requires — is not the branch AMBER is in.

The effect does not vanish over AMBER's available range of n. In a separate subsampling sweep at lower repetition count — which is why its full-size level differs slightly from Table 15's — the CCRC abort rate is 11.0% at $n = 228$, 10.2% at 190, 9.2% at 152 and 61.9% at 114: what binds is the interaction of fold size with the grid start rather than n alone. And the same probe run on POPE confirms that AMBER is not special: subsampled to $n = 228$, CCRC loses 3.65 points on POPE too, with abort rates of 43.5% (filtering) and 53.8% (CCRC). The setting that loses is the setting with the smallest calibration fold, on either dataset.

What this changes, and what it costs us. It costs us our only claimed instance of Conjecture 1, which is why that statement is now labelled a conjecture with no confirming evidence. It also costs us the clean story in which a checkable precondition screens deployments in advance. What replaces it is a different check, also computable before calibration and considerably better supported: given n_{cal}, α and δ, compute $k_{\min}$ and the emitted count at the grid start, ask how many errors the bound tolerates there, and ask whether the score's upper tail is clean enough to supply that. If the answer is "zero errors tolerated" — which it is for any $n_{\text{cal}} \lesssim 500$ at $\alpha = \delta = 0.10$ — then *any* perturbation of the strictest hypothesis, including a beneficial one, is a coin flip on the whole deployment. The practical remedies are equally elementary: a larger calibration fold, a looser α, or a grid start chosen so the first hypothesis tolerates at least one error. We flag rather than implement the last, because choosing it from data would need its own share of δ.

7 A sequential failure mode

Correction is more attractive in free-form generation than in yes/no answering: replacing a hallucinated word preserves a caption that deletion would mutilate. The extension looks mechanical — decompose the caption into claims, repair the unsupported ones, continue decoding — and two obstacles are already known. Editing a prefix makes later tokens off-policy, which is feedback covariate shift rather than exogenous shift (Fannjiang et al., 2022), so weighted conformal methods (Tibshirani et al., 2019) do not apply. And per-claim guarantees say nothing about a whole sequence, whose length is itself data-dependent.

Both could in principle be handled. Prefix-conditional (Mondrian) p-values make each decision valid given the realised prefix, so how we arrived there — including by our own edits — is absorbed into the conditioning; e-processes with Ville's inequality give sequence-level anytime validity, and are robust to adaptive intervention because our corrections are predictable with respect to the decoding filtration (Waudby-Smith & Ramdas, 2024; Ramdas et al., 2023). The residual difficulty is that the calibration pool drifts, since the deployed policy writes different text. The natural fix is on-policy calibration by iteration to a fixed point, and its convergence argument requires the correction map to be monotone. We measured whether it is, before attempting a proof.

Design. For each image we generate a caption greedily, locate the first hallucinated object mention at position t using AMBER's per-image present/absent object lists, and construct two prefixes *identical up to t*: one retaining the hallucinated word, one substituting the best-detected object the image genuinely contains. We then continue decoding the same number of tokens from each prefix and count hallucinated mentions in the continuations only. Matched prefixes make the contrast causal: the only difference between the two continuations is the intervention.

Result. Repair makes the continuation worse (Table 16, Figure 6). Downstream hallucinated mentions rise from 1.207 to 1.413, a paired difference of $+0.2066 \pm 0.0908$ ($n = 121$, $t(120) = 2.27$, two-sided $p = 0.025$), with 45 cases worsening against 25 improving and 51 tied. The sign is supported by three tests that make different assumptions — paired t $p = 0.025$, Wilcoxon signed-rank $p = 0.035$, exact sign test $p = 0.023$, all two-sided — which is the robustness evidence that matters given that 42% of pairs are tied. (Because of those ties the rank-based Hodges–Lehmann location estimate is exactly 0.000 even though the mean and every test statistic are positive; we state that rather than let a reader discover it.)

7.1 How much this experiment can support

The primary result is significant at conventional levels under three tests, and it is also under-powered. Both are true, and the second decides what may be concluded: the fixed-point route is not *closed* by this evidence, it is *unsupported* by it. The numbers, at the observed paired standard deviation of 0.9993 with $n = 121$:

- The two-sided minimum detectable effect at 80% power is $+0.2566$, which is *larger* than the observed $+0.2066$. The study was designed to detect effects of at least about a quarter of a hallucinated mention and detected a smaller one, which is a marginally-powered positive rather than a well-powered one.

Table 16: Paired matched-prefix test of monotonicity ($n = 121$ cases). Repair increases hallucination in the continuation. All p-values two-sided.

Downstream measure (continuation only)	Original	Repaired	paired diff. (p)
Hallucinated mentions (mean)	1.207	**1.413**	+0.207 (0.025)
Hallucinations per 100 words	3.61	4.30	+0.75 (0.011)
Faithful mentions (mean)	1.579	1.752	+0.174 (0.077)
Continuation length (words)	33.4	32.9	−0.55 (0.092)
Hallucinated share of mentions	0.433	0.447	−0.004 (0.884)
Primary endpoint: $+0.2066 \pm 0.0908$; $t(120) = 2.27$, $p = 0.025$; Wilcoxon $p = 0.035$; sign test $p = 0.023$			
95% CI $[+0.027, +0.386]$; Cohen's $d_z = 0.207$; Hodges–Lehmann +0.000			
Cases better / worse / tie: 25 / 45 / 51			

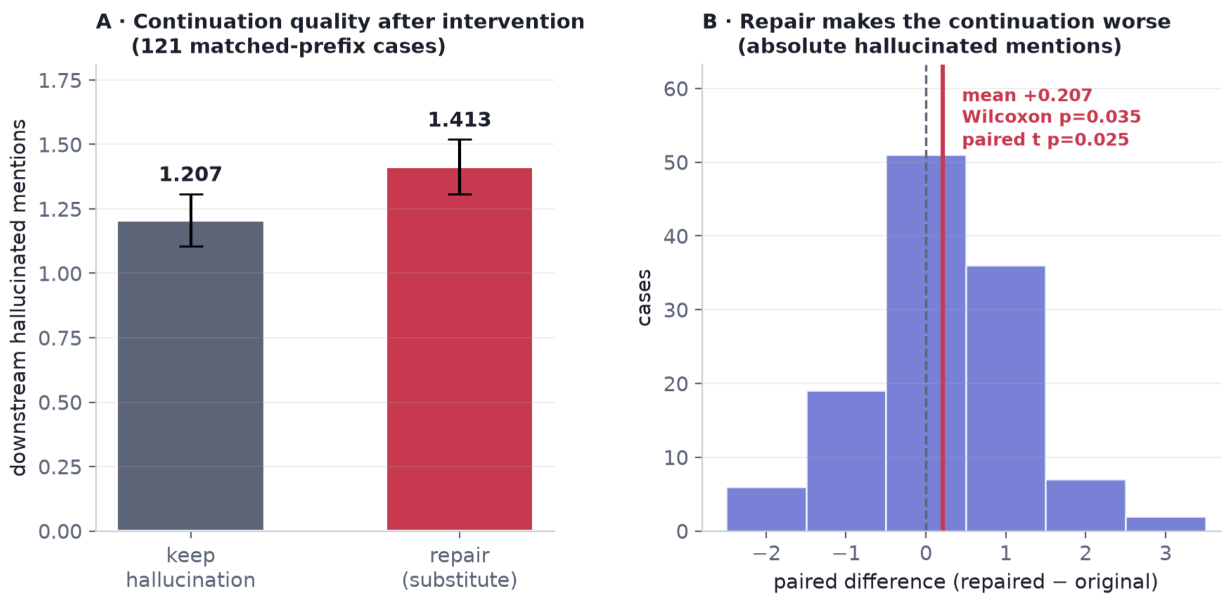

Figure 6: (A) Mean hallucinated mentions in the continuation after keeping versus repairing the flagged claim; error bars are standard errors. (B) Distribution of the paired difference; mass to the right of zero indicates repair worsening the continuation. The effect is modest in size and its sign is what matters for the theory; §7.1 bounds how little the experiment says about its magnitude.

- Achieved power at the observed effect is 61.6%; 186 pairs (65 more) would be needed for 80% power at this effect size, and 248 for 90%. (Post-hoc achieved power is a monotone restatement of the p-value and is not evidence about the true effect; we report it because it is the number most readers will want and the MDE is the one that carries information.)

- Equivalence testing bounds the effect from the other side. Against a pre-specified margin of ±0.25 — a quarter of a hallucinated object per repair, roughly 20% of the baseline — TOST does *not* establish equivalence ($p = 0.317$). The smallest margin we can establish at the 5% level is $\Delta = 0.357$, i.e. 29.6% of baseline.

- So the experiment simultaneously rules out an exact null, rules out true effects larger in magnitude than 0.357, and cannot distinguish +0.207 from anything in $[+0.027, +0.386]$ — a fourteen-fold range. **The increase is established as a sign, not as a magnitude.**

What survives is exactly what the theory needs and no more. A monotone fixed-point argument requires the correction map to be non-increasing in downstream risk; a sign is enough to withdraw support from that, and a magnitude would be required only to price the cost, which we therefore do not attempt to price. The right statement is: *our data does not support the monotonicity property a fixed-point calibration argument would rest on, and settling the question would take* $n \geq 186$ *matched pairs.* We do not claim the route is closed, and any downstream use of the number 0.207 as a quantity rather than as a direction is unsupported by this experiment.

We also pin down the per-word figure quoted in Table 16, since three defensible variants of it exist. The paired per-item rate difference over the 119 pairs with non-empty original continuations is +0.0075 per word (95% CI $[+0.0017, +0.0132]$, two-sided paired $p = 0.011$); the pooled ratio-of-totals is +0.0069 (paired-cluster bootstrap $p = 0.0086$). Including the two degenerate pairs whose original continuation is empty raises the estimate to +0.0115 and the p-value to 0.023, because one of them contributes +0.5 on a two-word continuation and supplies a third of the numerator. We report the 119-pair version, state the exclusion rule, and note that the result is not robust to those two pairs.

What this rules out, and what it does not. Because the correction map is not non-increasing on this evidence, monotone convergence arguments of Knaster–Tarski type have no basis here and we do not claim a sequential guarantee of that form. What survives is a coupling bound, which is weaker but assumption-light.

For a sequence of claims $O = (o_1, \dots, o_T)$ let $\mathcal{R}(O) = \mathbb{E}[\,\#\{t : o_t \text{ false}\}\,]$ be the expected count of false claims, the quantity a CHAIR-style loss charges for. Two regimes must be separated, and the distinction is exactly what §7 measures.

Theorem 2 (Coupling bound for corrected outputs)**.** *Let O denote the original output and O' the output after a repair rule is applied to a region R, and suppose the repair is* non-interfering*: conditional on not being repaired, every claim's truth value is unchanged by repairs elsewhere. Then*

$$\mathcal{R}(O') \;\leq\; \mathcal{R}(O) \;+\; \Pr[X \in R] \cdot \Pr[\textit{repair wrong} \mid X \in R].$$

Without non-interference the bound acquires a third term,

$$\mathcal{R}(O') \;\leq\; \mathcal{R}(O) \;+\; \Pr[X \in R] \cdot \Pr[\textit{repair wrong} \mid X \in R] \;+\; \mathcal{I}, \qquad \mathcal{I} \;:=\; \mathbb{E}\big[\Delta_{R^c}\big], \tag{4}$$

where Δ_{R^c} is the change in the number of false claims outside R induced by repairing inside it.

Proof. Decompose by whether a claim is repaired. Under non-interference the claims in R^c retain their truth values, contributing at most $\mathcal{R}(O)$; on R the emitted claim is wrong only if the repair is wrong, contributing at most $\Pr[X \in R]\Pr[\text{repair wrong} \mid X \in R]$. Summing gives the first bound. In general the R^c contribution is $\mathcal{R}(O) + \mathbb{E}[\Delta_{R^c}]$ by definition of Δ_{R^c}, which gives (4). ☐

Remark 7 (The interference term is not negligible here)*.* Non-interference is the assumption §7 falsifies: we measure $\widehat{\mathcal{I}} = +0.207 \pm 0.091$ additional false claims per repair ($n = 121$, $p = 0.025$), so for a single repair the interference term is roughly the same size as the direct repair-error term and cannot be dropped. Any procedure certifying corrected sequences must estimate $\mathcal{I}$ rather than assume it away, and $\mathcal{I}$ depends on the repair rule — which is why a context-aware rule is the natural next step rather than a tighter bound.

Remark 8 (What the corollary does and does not give)*.* Controlling $\mathcal{R}(O) \leq \alpha_1$ and the repair term by α_2 at confidence $1 - \delta/2$ each yields $\mathcal{R}(O') \leq \alpha_1 + \alpha_2 + \mathcal{I}$ at confidence $1 - \delta$. We flag a limitation of this composition rather than hiding it: certifying $\mathcal{R}(O)$ for the *uncorrected* generation is precisely the object the abstention floor makes expensive, so the corollary is useful only relative to an already-selective base policy (for which $\mathcal{R}$ is conditioned on emission), not to raw generation. Making the sequential guarantee self-contained — certifying the corrected sequence without first certifying the uncorrected one — remains open.

What this implies for algorithm design. The measurement identifies a cost that a myopic procedure does not charge for. Greedy per-claim repair optimises the risk of the claim in front of it while imposing an additional hallucinated claim on the continuation with probability we can sign but not size. A correct sequential procedure must therefore price the *continuation*, not the claim — for instance by searching over rewrite trajectories under an accumulated risk ledger rather than deciding each claim independently. §9 states why that is a genuinely different statistical object and why we regard it as the main open problem this paper leaves.

A competing explanation, and what survives it. Table 16 contains rows that complicate the headline. Repair raises faithful mentions by +0.174 ($p = 0.077$) at the same time as it raises hallucinated mentions by +0.207, in continuations that are slightly *shorter* (33.4 → 32.9 words, $p = 0.092$). Repaired prefixes therefore produce denser object talk — mention density rises about 15% in relative terms — and the obvious alternative reading is that correction makes the continuation more descriptive rather than less faithful. Two normalisations settle it. Per word, hallucination rises significantly (+0.0075 per word, two-sided $p = 0.011$). Per *mention*, it does not move at all: the hallucinated share of mentions goes 0.433 → 0.447 pooled, and the paired difference is -0.0039 ($p = 0.884$, $n = 111$ pairs with mentions in both arms). The competing explanation is therefore substantially correct as a description of *why* the count rises.

It does not, however, rescue the monotonicity argument, and the reason is that the quantity being certified is a count and not a ratio. A CHAIR-style risk charges for each hallucinated mention emitted; a procedure that repairs one claim and emits two additional false ones has increased the risk of its own output regardless of how many true claims came with them. What the normalised results do establish is the mechanism — repair perturbs the generation towards denser description, and hallucination scales with density — and that a density-controlled repair rule might avoid the cost. We report the share result because a reader is entitled to it, and because it makes the negative result narrower than the raw count suggests.

Scope of the negative result. Our repair substitutes the globally best-detected object, which can be contextually inappropriate mid-sentence and may push the prefix off-policy for reasons unrelated to faithfulness. The supported claim is that *context-blind* substitution carries a downstream cost in absolute and per-word terms, not that all correction must, and not that it degrades the faithful/hallucinated *balance*. Constraining repairs to the model's own top-k continuations — which the one-shot procedure of §4 already implies — is the natural remedy and remains to be tested.

8 Ablations

8.1 The score combiner, and a validity trap

Channel 1 is a learned function of a handful of scalars. It is natural to ask whether a more expressive combiner helps. It does not, and it can silently invalidate the guarantee.

Table 17 reports two protocols, and the audit statistic is the exceedance fraction rather than mean risk. Under the naive protocol, where the combiner is fitted on the same fold used to calibrate the threshold, gradient boosting and random forests appear to certify far more coverage — 89.2% and 86.1% — but 93.5% and 72.5% of their splits *exceed* the risk target on held-out data. Their mean realised risks, 13.3% and 12.1%, badly understate that: the exceedance fraction is the guarantee actually being broken. The mechanism is straightforward: a high-capacity model overfits the calibration scores, so empirical error on the calibration fold understates true error and the threshold is set too permissively. (Note that under this protocol the full-item-set exceedance is *not* a valid proxy — two thirds of those items were used to fit the score, so random forest looks fine at 6.8% there while failing at 72.5% on held-out data. Where fit and calibration overlap, only the held-out column may be read.)

Under a three-way split all combiners are valid — but they are *not* tied. Logistic regression beats gradient boosting by 11.90 points ($p = 8 \times 10^{-18}$) and random forest by 15.41 points ($p = 2 \times 10^{-20}$), and the reason is again the abort rate: 14.5% and 24.8% of splits against logistic regression's 0.0%. A high-capacity score here is not merely no better; it fails the first hypothesis of the fixed sequence on up to a quarter of splits. The MLP is worse still, 14.2% mean coverage with a 60.8% abort rate, from overfitting six features.

We draw two conclusions. Practically, the score must be fitted on a fold disjoint from calibration — a discipline that costs 3.6 points of coverage for logistic regression and 30 points for the tree ensembles, and is

Table 17: Combiner ablation on POPE-1500 ($\alpha = \delta = 0.10$, 400 paired repetitions, six features). The audit column is Pr[realised risk $> \alpha$], not mean risk. Under data reuse, high-capacity combiners *break* the guarantee on the majority of splits; under a three-way split they are valid but abort. Logistic regression is chosen deliberately, and the abort column is the reason.

	Naive: fit = calibrate		Three-way split			
Combiner	cov@10%	$\widehat{\Pr}[\mathcal{R} > \alpha]$ (test fold)	cov@10%	abort	$\widehat{\Pr}[\mathcal{R} > \alpha]$ (all items)	vs. logistic
Logistic regression	74.3%	18.5%	**70.7%**	**0.0%**	3.2%	—
Gradient boosting	89.2%	**93.5%** (violated)	58.8%	14.5%	0.0%	−11.90
Random forest	86.1%	**72.5%** (violated)	55.3%	24.8%	0.3%	−15.41
MLP $(64, 32)$	25.6%	14.9%	14.2%	60.8%	1.3%	−56.49

Table 18: Risk-bound ablation (POPE-1500, three-way split, 400 repetitions). Exact binomial dominates for a binary loss; the others are valid but conservative. $k_{\min}$ is the smallest emitted set the bound can certify with zero observed errors (Lemma 1), which is a pre-calibration screening quantity: a bound whose $k_{\min}$ exceeds what the grid start can emit certifies nothing. The audit column is the fraction of splits whose risk on all items exceeds α, against $\delta = 0.10$.

		$\alpha = 0.10$			$\alpha = 0.15$		
Upper confidence bound	$k_{\min}$	cov	risk	$\widehat{\Pr}[\mathcal{R} > \alpha]$	cov	risk	$\widehat{\Pr}[\mathcal{R} > \alpha]$
Hoeffding	116	18.8%	4.2%	0%	75.8%	9.4%	0%
Empirical Bernstein	71	23.2%	3.2%	0%	70.3%	8.8%	0%
Betting (e-CRC)	44	62.5%	5.9%	0%	78.1%	9.9%	0%
Clopper–Pearson (exact)	22	**70.7%**	7.8%	3%	**86.1%**	12.5%	2%

necessary for any scorer whose capacity is not negligible. Conceptually, capacity is not the bottleneck: the features are already informative scalars, and what limits efficiency is the quality of the grounding signal, not the flexibility of the combiner — with the addendum that excess capacity is not free but actively harmful under a valid sequential test.

8.2 The risk bound

The emitted-set error is a binomial proportion, so an exact interval should dominate general-purpose concentration inequalities, and it does: at $\alpha = 0.10$ Clopper–Pearson certifies 70.7% coverage, the betting bound 62.5%, empirical Bernstein 23.2% and Hoeffding 18.8% (Table 18). All four are valid; the loose two are simply conservative, realising 3.2% and 4.2% risk against a 10% budget. This is consistent with the hierarchy established by Kotte (2026a) and indicates that for binary losses the choice of bound is worth more than fifty points of coverage. The $k_{\min}$ column is the same story from the other end and is the one to check before choosing: at $\alpha = \delta = 0.10$ Hoeffding cannot certify an emitted set smaller than 116 items, so on a 76-item calibration fold it cannot certify anything at all whatever the score does. Variance-adaptive or betting-style bounds (Waudby-Smith & Ramdas, 2024) become preferable for graded losses, which is the natural setting for claim-level factuality scores.

8.3 Positioning summary

Figure 7 summarises where CCRC sits. Panel A gives risk–coverage curves; panel B contrasts capabilities across recent methods. Mitigation methods (Huang et al., 2024; Jiang et al., 2025; Wu et al., 2025) operate only at full coverage and offer no guarantee; conformal methods (Li et al., 2025; Lee et al., 2025) guarantee but only delete or abstain; Xu et al. (2026) re-scores without changing the answer. CCRC is, to our knowledge, the first to certify an output that differs from the model's — while requiring, as §4.4 shows, an independent channel to do so in practice.

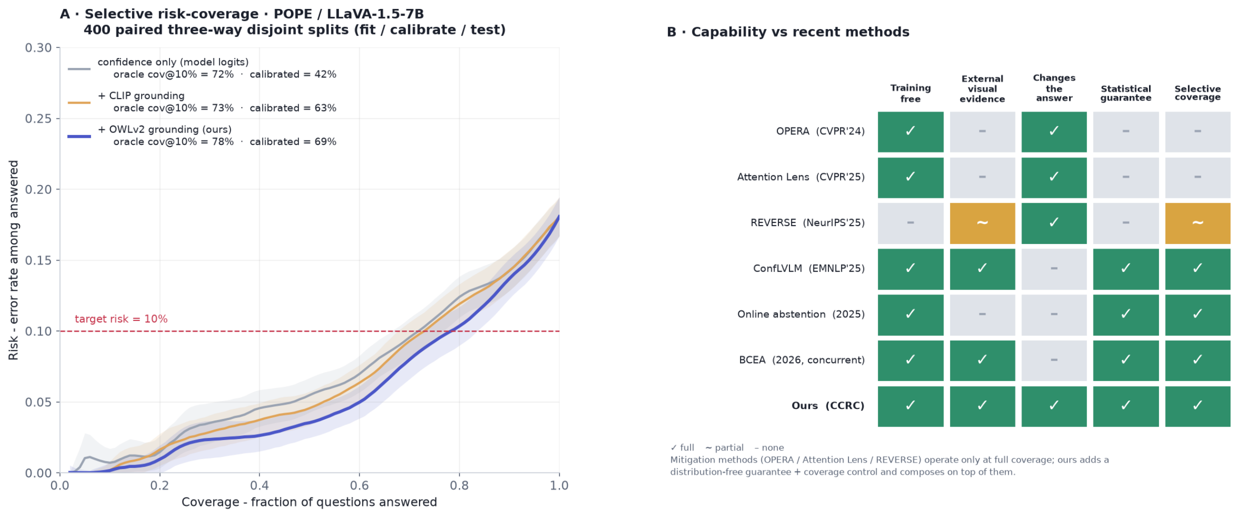

Figure 7: (A) Risk–coverage on POPE-1500 for the three score variants, 400 paired three-way splits, with oracle and conformally calibrated coverage at $\alpha = 0.10$ labelled separately in the legend. (B) Capability comparison with recent methods. Mitigation approaches change the base model but provide no guarantee; existing conformal approaches guarantee but only remove or abstain; Xu et al. (2026) guarantees and acquires evidence but does not change the emitted answer. Our layer composes on top of mitigation decoders (§5.5).

9 Discussion

What the results support. Permitting a selective predictor to emit a substituted answer is a real degree of freedom: it evades the premise of Proposition 1, the guarantee survives the substitution (Theorem 1) because risk control cares about error indicators rather than about provenance, and in eight of ten cells it converts a repaired mass of 1–2% into +0.4 to +7.6 points of certified coverage with intervals excluding zero. The mechanism is understood well enough to be decomposed: most of that is the repaired items being emitted at almost no risk cost, and +0.2 to +1.8 points is genuine dilution.

What they do not. The gains are modest and conditional. The mechanism requires a strict gate (Table 5), an independent channel (§4.4), and a calibration fold large enough that the first hypothesis of the fixed sequence can pass (§6) — and the last of these costs us more on the setting where it binds than the mechanism gains anywhere. What we do *not* have is a clean headroom precondition: Conjecture 1 has no confirming instance in our data. Most of our margin over a published baseline is the correctness score rather than the algorithm — 64.5 points against 2.3 (Table 13) — and the closest concurrent method is complementary rather than dominated (Table 14). The one-shot theory is a correct but standard application of fixed-sequence testing with exact bounds, and it is stated for a value-indexed family while our measurements come from the rank-indexed variant (Remark 3). The interesting theory is sequential, and §7 shows that we cannot currently support the most natural route to it.

Limitations. Channel 2 is an object detector, so the evaluated method addresses existence hallucination — and, within that, only *missed* objects: §5.3 shows that even a polarity-symmetric gate corrects no false-positive existence hallucination in any setting, because on the negative side the reachable items are the ones the model already gets right. On attribute-, counting- and reasoning-heavy benchmarks the method degenerates to filtering, which remains valid but gains nothing. The independence of channel 2 is an assumption, and our own instantiation violates it (channel 1's features include the detector score); a detector sharing the VLM's blind spots would weaken certification without invalidating it. The repair gate q is fixed a priori rather than learned; §4.6 prices what that choice is worth and Appendix A.4 gives a fully valid alternative. Exchangeability is assumed throughout; under deployment shift the stated level need not hold, and adaptive conformal methods (Gibbs & Candès, 2021) would be required. Our sequential experiment uses one backbone, a context-blind repair rule, and $n = 121$ pairs, which is under-powered as a magnitude claim (§7.1). And every interval in this paper is over splits of a fixed dataset, so none of them speaks to replication on new items.

Outlook. The open problem we consider most interesting is the one the sequential negative result exposes. Because repairing a claim changes which subsequent claims exist, a sequential procedure faces an endogenous item set and must select among rewrite trajectories rather than filter a fixed list. Selecting the best trajectory and then reporting its certificate is a selective-inference error, so a correct procedure needs a certificate simultaneous over the search tree. Settling whether repair degrades continuations at all would take $n \geq 186$ matched pairs; building the simultaneous certificate is the larger task. A second, cheaper item: the failure mode of §6 suggests that fixed-sequence testing is the wrong search strategy at small calibration size, and that a method spending a little δ to avoid staking everything on the strictest hypothesis would dominate it exactly where our method currently loses.

10 Conclusion

Selective prediction for vision–language models is expensive because of a bound whose premise — that the emitted answer is the model's own — is an assumption about the action space rather than about the data. Relaxing it works, but not freely and not by much: certified correction requires an independent evidence channel, a strict repair gate, and a calibration fold large enough for the sequential test to get started, and what it delivers is a couple of points of certified coverage against the sixty-odd that the correctness score delivers. Applied myopically in generation, the same idea makes things measurably worse. We have tried to report the conditions and the failures as carefully as the improvements — including the ones that cost us a proposition, a multiplier and a diagnosis — in the belief that for methods whose selling point is a guarantee, the boundaries of that guarantee are the substance of the contribution.

Broader Impact Statement

Attaching guarantees to model outputs can improve safety, but a guarantee is only as meaningful as its assumptions. Our procedure controls the error rate among emitted answers under exchangeability and requires a genuinely independent evidence channel; if either fails — under deployment shift, or with a detector sharing the model's blind spots — the stated level may not hold. Emitting a *corrected* answer also changes the failure mode presented to a user, replacing an abstention that invites scrutiny with a confident substitution that may not. For this reason we regard the checkable precondition and the sequential negative result as safety-relevant components of the contribution rather than limitations to be minimised, and we recommend that any deployment verify the precondition and monitor channel independence rather than treating the nominal α as a guarantee that transfers unconditionally.

References

Anastasios N. Angelopoulos and Stephen Bates. Conformal prediction: A gentle introduction. *Foundations and Trends in Machine Learning*, 16(4):494–591, 2023.

Anastasios N. Angelopoulos, Stephen Bates, Emmanuel J. Candès, Michael I. Jordan, and Lihua Lei. Learn then test: Calibrating predictive algorithms to achieve risk control. *arXiv preprint arXiv:2110.01052*, 2021.

Anastasios N. Angelopoulos, Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster. Conformal risk control. In *International Conference on Learning Representations (ICLR)*, 2024.

Zechen Bai, Pichao Wang, Tianjun Xiao, Tong He, Zongbo Han, Zheng Zhang, and Mike Zheng Shou. Hallucination of multimodal large language models: A survey. *arXiv preprint arXiv:2404.18930*, 2024.

Rina Foygel Barber, Emmanuel J. Candès, Aaditya Ramdas, and Ryan J. Tibshirani. Predictive inference with the jackknife+. *Annals of Statistics*, 49(1):486–507, 2021.

Rina Foygel Barber, Emmanuel J. Candès, Aaditya Ramdas, and Ryan J. Tibshirani. Conformal prediction beyond exchangeability. *Annals of Statistics*, 51(2):816–845, 2023.

Peter L. Bartlett and Marten H. Wegkamp. Classification with a reject option using a hinge loss. *Journal of Machine Learning Research*, 9:1823–1840, 2008.

Stephen Bates, Anastasios N. Angelopoulos, Lihua Lei, Jitendra Malik, and Michael I. Jordan. Distribution-free, risk-controlling prediction sets. *Journal of the ACM*, 68(6):43:1–43:34, 2021.

Lingjiao Chen, Matei Zaharia, and James Zou. Frugalgpt: How to use large language models while reducing cost and improving performance. *Transactions on Machine Learning Research*, 2024a. URL `https://arxiv.org/abs/2305.05176`.

Yuzhu Chen, Yingjie Wang, Shunyu Liu, Yongcheng Jing, and Dacheng Tao. CoVeR: Conformal calibration for versatile and reliable autoregressive next-token prediction. *arXiv preprint arXiv:2509.04733*, 2025.

Zhaorun Chen, Zhuokai Zhao, Hongyin Luo, Huaxiu Yao, Bo Li, and Jiawei Zhou. Halc: Object hallucination reduction via adaptive focal-contrast decoding. In *International Conference on Machine Learning (ICML)*, 2024b.

Zhe Cheng, Wenyu Chen, Fode Zhang, and Dehuan Shen. Mitigating hallucinations in large vision-language models via causal route gating. In *International Conference on Machine Learning (ICML)*, 2026. URL `https://arxiv.org/abs/2605.24024`. Spotlight.

C. K. Chow. On optimum recognition error and reject tradeoff. *IEEE Transactions on Information Theory*, 16(1):41–46, 1970.

C. J. Clopper and E. S. Pearson. The use of confidence or fiducial limits illustrated in the case of the binomial. *Biometrika*, 26(4):404–413, 1934.

Jeremy M. Cohen, Elan Rosenfeld, and J. Zico Kolter. Certified adversarial robustness via randomized smoothing. In *International Conference on Machine Learning (ICML)*, 2019.

Corinna Cortes, Giulia DeSalvo, and Mehryar Mohri. Learning with rejection. In *International Conference on Algorithmic Learning Theory (ALT)*, pp. 67–82, 2016.

Quang-Vinh Dang, Ngoc-Son-An Nguyen, Thi-Bich-Diem Vo, Minh Ngoc Dinh, Anh Dat Le, Hoang Viet Vu, and Phuong Lan Nguyen. Real-time hallucination correction in vision-language models using dynamic knowledge graph verification. *Discover Artificial Intelligence*, 2026. In press.

Jingyuan Deng and Yujiu Yang. MaskCD: Mitigating LVLM hallucinations by image head masked contrastive decoding. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pp. 18854–18866. Association for Computational Linguistics, 2025. URL `https://arxiv.org/abs/2510.02790`.

Tiffany Ding, Anastasios N. Angelopoulos, Stephen Bates, Michael I. Jordan, and Ryan J. Tibshirani. Class-conditional conformal prediction with many classes. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Ran El-Yaniv and Yair Wiener. On the foundations of noise-free selective classification. *Journal of Machine Learning Research*, 11:1605–1641, 2010.

Clara Fannjiang, Stephen Bates, Anastasios N. Angelopoulos, Jennifer Listgarten, and Michael I. Jordan. Conformal prediction under feedback covariate shift for biomolecular design. *Proceedings of the National Academy of Sciences*, 119(43), 2022.

Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630:625–630, 2024.

Alessandro Favero, Luca Zancato, Matthew Trager, Siddharth Choudhary, Pramuditha Perera, Alessandro Achille, Ashwin Swaminathan, and Stefano Soatto. Multi-modal hallucination control by visual information grounding. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

Chaoyou Fu, Peixian Chen, Yunhang Shen, Yulei Qin, Mengdan Zhang, Xu Lin, Jinrui Yang, Xiawu Zheng, Ke Li, Xing Sun, Yunsheng Wu, Rongrong Ji, Caifeng Shan, and Ran He. MME: A comprehensive evaluation benchmark for multimodal large language models. *arXiv preprint arXiv:2306.13394*, 2023. URL `https://arxiv.org/abs/2306.13394`.

Yonatan Geifman and Ran El-Yaniv. Selectivenet: A deep neural network with an integrated reject option. In *International Conference on Machine Learning (ICML)*, 2019.

Isaac Gibbs and Emmanuel J. Candès. Adaptive conformal inference under distribution shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2021.

Tianrui Guan, Fuxiao Liu, Xiyang Wu, Ruiqi Xian, Zongxia Li, Xiaoyu Liu, Xijun Wang, Lichang Chen, Furong Huang, Yaser Yacoob, Dinesh Manocha, and Tianyi Zhou. HallusionBench: An advanced diagnostic suite for entangled language hallucination and visual illusion in large vision-language models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

Yu Gui, Ying Jin, and Zhimei Ren. Conformal alignment: Knowing when to trust foundation models with guarantees. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *International Conference on Machine Learning (ICML)*, 2017.

Neha Gupta, Harikrishna Narasimhan, Wittawat Jitkrittum, Ankit Singh Rawat, Aditya Krishna Menon, and Sanjiv Kumar. Language model cascades: Token-level uncertainty and beyond. In *International Conference on Learning Representations (ICLR)*, 2024.

Wassily Hoeffding. On the distribution of the number of successes in independent trials. *Annals of Mathematical Statistics*, 27(3):713–721, 1956.

Qidong Huang, Xiaoyi Dong, Pan Zhang, Bin Wang, Conghui He, Jiaqi Wang, Dahua Lin, Weiming Zhang, and Nenghai Yu. OPERA: Alleviating hallucination in multi-modal large language models via over-trust penalty and retrospection-allocation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Seattle, WA, USA, 2024.

Drew A. Hudson and Christopher D. Manning. GQA: A new dataset for real-world visual reasoning and compositional question answering. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.

Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Yejin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55 (12):248:1–248:38, 2023.

Zhangqi Jiang, Junkai Chen, Beier Zhu, Tingjin Luo, Yankun Shen, and Xu Yang. Devils in middle layers of large vision-language models: Interpreting, detecting and mitigating object hallucinations via attention lens. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025. URL `https://arxiv.org/abs/2411.16724`.

Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.

Varun Kotte. When can conformal risk control certify LLM outputs? bounds, impossibility, and adaptation for structured generation. *arXiv preprint arXiv:2606.29054*, 2026a. URL `https://arxiv.org/abs/2606.29054`.

Varun Kotte. PASC: Pipeline-aware conformal prediction with joint coverage guarantees for multi-stage NLP and LLM pipelines. *arXiv preprint arXiv:2605.18812*, 2026b.

Lorenz Kuhn, Yarin Gal, and Sebastian Farquhar. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *International Conference on Learning Representations (ICLR)*, 2023.

Bhawesh Kumar, Charlie Lu, Gauri Gupta, Anil Palepu, David Bellamy, Ramesh Raskar, and Andrew Beam. Conformal prediction with large language models for multi-choice question answering. In *ICML 2023 Workshop on Neural Conversational AI: What's Left to TEACH (Trustworthy, Enhanced, Adaptable, Capable and Human-centric) Chatbots?*, 2023. URL `https://arxiv.org/abs/2305.18404`.

Claire Le Goues, Michael Pradel, and Abhik Roychoudhury. Automated program repair. *Communications of the ACM*, 62(12):56–65, 2019.

Junlin Lee, Yequan Wang, Jing Li, and Min Zhang. Multimodal reasoning with multimodal knowledge graph. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, pp. 10767–10782, Bangkok, Thailand, 2024. URL `https://aclanthology.org/2024.acl-long.579/`.

Minjae Lee, Yoonjae Jung, and Sangdon Park. Online conformal abstention for factuality control under adversarial bandit feedback. *arXiv preprint arXiv:2506.14067*, 2025.

Jing Lei, Max G'Sell, Alessandro Rinaldo, Ryan J. Tibshirani, and Larry Wasserman. Distribution-free predictive inference for regression. *Journal of the American Statistical Association*, 113(523):1094–1111, 2018.

Sicong Leng, Hang Zhang, Guanzheng Chen, Xin Li, Shijian Lu, Chunyan Miao, and Lidong Bing. Mitigating object hallucinations in large vision-language models through visual contrastive decoding. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Seattle, WA, USA, 2024.

Yifan Li, Yifan Du, Kun Zhou, Jinpeng Wang, Wayne Xin Zhao, and Ji-Rong Wen. Evaluating object hallucination in large vision-language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 292–305, Singapore, 2023. Association for Computational Linguistics.

Zhuohang Li, Chao Yan, Nicholas J. Jackson, Wendi Cui, Bo Li, Jiaxin Zhang, and Bradley A. Malin. Towards statistical factuality guarantee for large vision–language models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, Suzhou, China, 2025. Association for Computational Linguistics. URL `https://aclanthology.org/2025.emnlp-main.576/`.

Stephanie Lin, Jacob Hilton, and Owain Evans. Teaching models to express their uncertainty in words. *Transactions on Machine Learning Research*, 2022.

Fuxiao Liu, Kevin Lin, Linjie Li, Jianfeng Wang, Yaser Yacoob, and Lijuan Wang. Mitigating hallucination in large multi-modal models via robust instruction tuning. In *International Conference on Learning Representations (ICLR)*, 2024a.

Hanchao Liu, Wenyuan Xue, Yifei Chen, Dapeng Chen, Xiutian Zhao, Ke Wang, Liping Hou, Rongjun Li, and Wei Peng. A survey on hallucination in large vision–language models. *arXiv preprint arXiv:2402.00253*, 2024b.

Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Seattle, WA, USA, 2024c.

Shilong Liu, Zhaoyang Zeng, Tianhe Ren, Feng Li, Hao Zhang, Jie Yang, Qing Jiang, Chunyuan Li, Jianwei Yang, Hang Su, Jun Zhu, and Lei Zhang. Grounding dino: Marrying dino with grounded pre-training for open-set object detection. In *European Conference on Computer Vision (ECCV)*, 2024d.

Potsawee Manakul, Adian Liusie, and Mark J. F. Gales. Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models. In *Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

Matthias Minderer, Alexey A. Gritsenko, and Neil Houlsby. Scaling open-vocabulary object detection. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.

Christopher Mohri and Tatsunori Hashimoto. Language models with conformal factuality guarantees. In *International Conference on Machine Learning (ICML)*, 2024. arXiv:2402.10978.

Lingyou Pang, Lei Huang, Jianyu Lin, Tianyu Wang, Alexander Aue, and Carey E. Priebe. Taming variability: Randomized and bootstrapped conformal risk control for LLMs. *arXiv preprint arXiv:2509.23007*, 2025.

Harris Papadopoulos, Kostas Proedrou, Volodya Vovk, and Alex Gammerman. Inductive confidence machines for regression. In *European Conference on Machine Learning (ECML)*, pp. 345–356, 2002.

Drew Prinster, Samuel Stanton, Anqi Liu, and Suchi Saria. Conformal validity guarantees exist for any data distribution (and how to find them). In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pp. 41086–41118. PMLR, 2024. URL `https://arxiv.org/abs/2405.06627`.

Victor Quach, Adam Fisch, Tal Schuster, Adam Yala, Jae Ho Sohn, Tommi S. Jaakkola, and Regina Barzilay. Conformal language modeling. In *International Conference on Learning Representations (ICLR)*, 2024. URL `https://arxiv.org/abs/2306.10193`.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning (ICML)*, 2021.

Aaditya Ramdas, Peter Grünwald, Vladimir Vovk, and Glenn Shafer. Game-theoretic statistics and safe anytime-valid inference. *Statistical Science*, 38(4):576–601, 2023.

Allen Z. Ren, Anushri Dixit, Alexandra Bodrova, Sumeet Singh, Stephen Tu, Noah Brown, Peng Xu, Leila Takayama, Fei Xia, Jake Varley, Zhenjia Xu, Dorsa Sadigh, Andy Zeng, and Anirudha Majumdar. Robots that ask for help: Uncertainty alignment for large language model planners. In *Conference on Robot Learning (CoRL)*, 2023.

Anna Rohrbach, Lisa Anne Hendricks, Kaylee Burns, Trevor Darrell, and Kate Saenko. Object hallucination in image captioning. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 4035–4045, Brussels, Belgium, 2018. Association for Computational Linguistics.

Yaniv Romano, Evan Patterson, and Emmanuel J. Candès. Conformalized quantile regression. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

Mauricio Sadinle, Jing Lei, and Larry Wasserman. Least ambiguous set-valued classifiers with bounded error levels. *Journal of the American Statistical Association*, 114(525):223–234, 2019.

Zhiqing Sun, Sheng Shen, Shengcao Cao, Haotian Liu, Chunyuan Li, Yikang Shen, Chuang Gan, Liangyan Gui, Yu-Xiong Wang, Yiming Yang, Kurt Keutzer, and Trevor Darrell. Aligning large multimodal models with factually augmented RLHF. In *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 13088–13110, Bangkok, Thailand, 2024. Association for Computational Linguistics. URL `https://aclanthology.org/2024.findings-acl.775/`.

Ryan J. Tibshirani, Rina Foygel Barber, Emmanuel J. Candès, and Aaditya Ramdas. Conformal prediction under covariate shift. In *Advances in Neural Information Processing Systems (NeurIPS)*, pp. 2526–2536, 2019.

Vladimir Vovk, David Lindsay, Ilia Nouretdinov, and Alex Gammerman. Mondrian confidence machine. On-line Compression Modelling project, Working Paper 4, Royal Holloway, University of London, March 2003.

Vladimir Vovk, Alexander Gammerman, and Glenn Shafer. *Algorithmic Learning in a Random World*. Springer, New York, 2005.

Benedikt J. Wagner. Two axes of LLM abstention: Answer correctness and question answerability. *arXiv preprint arXiv:2607.08456*, 2026.

Zifu Wan, Ce Zhang, Silong Yong, Martin Q. Ma, Simon Stepputtis, Louis-Philippe Morency, Deva Ramanan, Katia Sycara, and Yaqi Xie. ONLY: One-layer intervention sufficiently mitigates hallucinations in large vision-language models. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 3225–3234, 2025. URL `https://arxiv.org/abs/2507.00898`.

Junyang Wang, Yuhang Wang, Guohai Xu, Jing Zhang, Yukai Gu, Haitao Jia, Jiaqi Wang, Haiyang Xu, Ming Yan, Ji Zhang, and Jitao Sang. AMBER: An LLM-free multi-dimensional benchmark for MLLMs hallucination evaluation. *arXiv preprint arXiv:2311.07397*, 2023a. URL `https://arxiv.org/abs/2311.07397`.

Peng Wang, Shuai Bai, Sinan Tan, Shijie Wang, Zhihao Fan, Jinze Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Yang Fan, Kai Dang, Mengfei Du, Xuancheng Ren, Rui Men, Dayiheng Liu, Chang Zhou, Jingren Zhou, and Junyang Lin. Qwen2-VL: Enhancing vision-language model's perception of the world at any resolution. *arXiv preprint arXiv:2409.12191*, 2024.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*, 2023b.

Ian Waudby-Smith and Aaditya Ramdas. Estimating means of bounded random variables by betting. *Journal of the Royal Statistical Society Series B*, 86(1):1–27, 2024.

Tsung-Han Wu, Heekyung Lee, Jiaxin Ge, Joseph E. Gonzalez, Trevor Darrell, and David M. Chan. Generate, but verify: Reducing hallucination in vision-language models with retrospective resampling. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. URL `https://reverse-vlm.github.io/`.

Peng Xia, Kangyu Zhu, Haoran Li, Hongtu Zhu, Yun Li, Gang Li, Linjun Zhang, and Huaxiu Yao. RULE: Reliable multimodal RAG for factuality in medical vision language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 1081–1093, Miami, Florida, USA, 2024.

Beining Xu and Yongming Lu. TECP: Token-entropy conformal prediction for LLMs. *Mathematics*, 13(20): 3351, 2025. URL `https://doi.org/10.3390/math13203351`.

Jian Xu, Yanning Wu, Delu Zeng, John Paisley, and Qibin Zhao. Look again before you abstain: Budgeted conformal evidence acquisition for reliable vision–language models. *arXiv preprint arXiv:2606.16667*, 2026. URL `https://arxiv.org/abs/2606.16667`.

Shukang Yin, Chaoyou Fu, Sirui Zhao, Tong Xu, Hao Wang, Dianbo Sui, Yunhang Shen, Ke Li, Xing Sun, and Enhong Chen. Woodpecker: Hallucination correction for multimodal large language models. *Science China Information Sciences*, 67(12):220105:1–220105:13, 2024. URL `https://doi.org/10.1007/s11432-024-4251-x`.

Tianyu Yu, Yuan Yao, Haoye Zhang, Taiwen He, Yifeng Han, Ganqu Cui, Jinyi Hu, Zhiyuan Liu, Hai-Tao Zheng, Maosong Sun, and Tat-Seng Chua. Rlhf-v: Towards trustworthy mllms via behavior alignment from fine-grained correctional human feedback. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.

Ce Zhang, Zifu Wan, Zhehan Kan, Martin Q. Ma, Simon Stepputtis, Deva Ramanan, Russ Salakhutdinov, Louis-Philippe Morency, Katia Sycara, and Yaqi Xie. Self-correcting decoding with generative feedback for mitigating hallucinations in large vision-language models. In *International Conference on Learning Representations (ICLR)*, 2025. URL `https://arxiv.org/abs/2502.06130`.

Linxi Zhao, Yihe Deng, Weitong Zhang, and Quanquan Gu. Mitigating object hallucination in large vision-language models via image-grounded guidance. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pp. 77461–77486. PMLR, 2025a. URL `https://proceedings.mlr.press/v267/zhao25j.html`. Spotlight.

Qiyan Zhao, Xiaofeng Zhang, Yiheng Li, Yun Xing, Xiaosong Yuan, Feilong Tang, Sinan Fan, Xuhang Chen, Xuyao Zhang, and Dahan Wang. MCA-LLaVA: Manhattan causal attention for reducing hallucination in large vision-language models. In *Proceedings of the 33rd ACM International Conference on Multimedia (ACM MM)*. Association for Computing Machinery, 2025b. URL `https://arxiv.org/abs/2507.09184`.

Yiyang Zhou, Chenhang Cui, Jaehong Yoon, Linjun Zhang, Zhun Deng, Chelsea Finn, Mohit Bansal, and Huaxiu Yao. Analyzing and mitigating object hallucination in large vision-language models. In *International Conference on Learning Representations (ICLR)*, Vienna, Austria, 2024.

A Additional results

A.1 Synthetic validation of the three pillars

Before any real-data experiment we validated the procedure on synthetic data where ground truth is controllable, confirming (i) that the certified risk is respected, (ii) that a better-separated score buys coverage at fixed risk, and (iii) that a single global threshold can satisfy a marginal guarantee while violating it on a minority subgroup (25.3% realised risk against a 10% target), which group-conditional (Mondrian) calibration repairs (9.8% and 9.7% on the two groups). The last point motivates treating conditional coverage as a separate requirement from marginal coverage; we report it on synthetic data only, as our real-data samples are not category-balanced.

A second synthetic study is the one Remark 3 relies on: it fixes a known $\Pr[Y = 1 \mid s]$ so that the population risk of any selected threshold can be computed by quadrature rather than estimated, and measures the exceedance of the *rank*-indexed fixed sequence — the variant we implement, and the one the theorem does not cover — over four score shapes × four calibration sizes at 4000 replications per cell. Unconditional exceedance is 0.0–5.9% against $\delta = 0.10$ in all sixteen cells. The simulation also reproduces §6 from first principles: abort rates run from 1% to 100% across the grid and are governed by how fast $\Pr[Y = 1 \mid s]$ rises in the extreme upper tail, not by the score's average separability.

A.2 Multi-α table

Table 19 gives certified coverage at four risk levels, the numerical version of Figure 4. The abort block is the part worth reading: at $\alpha = 0.05$, $k_{\min} = 45$ and two thirds of splits certify nothing on the confidence-only score.

A.3 Validity audit, all cells

Table 20 is the audit referred to throughout: the fraction of splits whose realised risk exceeds α, against $\delta = 0.10$, for every setting × risk level × arm. The column that decides the question is `exc(all)`, the selected policy re-scored on all n items, which is a low-noise proxy for its population risk; `exc(te)` is the same statistic on the $n/3$ held-out fold and is inflated by binomial noise at that size. Aborted splits emit nothing and are excluded from the denominators.

Table 19: Certified coverage and abort rate at four risk targets (POPE-1500, LLaVA-1.5, 400 paired splits). $k_{\min}$ is 45, 22, 15, 11 at the four targets. Gains over confidence-only are paired; all exclude zero except CLIP at $\alpha = 0.20$ (-0.06, $p = 0.27$), which is why "the grounded score dominates at every level" is stated in §5.1 for detection grounding only.

Score	$\alpha = 0.05$	0.10	0.15	0.20
certified coverage				
Confidence only	9.0%	41.6%	81.4%	97.3%
+ CLIP grounding	13.9%	63.4%	82.5%	97.2%
+ detection grounding	**27.3%**	**68.6%**	**86.4%**	**97.7%**
abort rate				
Confidence only	65.8%	32.5%	0.0%	0.0%
+ CLIP grounding	49.8%	0.5%	0.0%	0.0%
+ detection grounding	**24.0%**	**0.8%**	0.0%	0.0%
paired gain of detection grounding over confidence only				
	+18.29	+27.01	+5.03	+0.48

Table 20: Validity audit over all 40 cells (400 paired splits each). Every cell satisfies `exc(all)` $\leq \delta = 0.10$. Ranges: `exc(all)` 0.0–5.1%, `exc(te)` 3.1–22.3%.

		filtering		$q = 0.10$		$q = 0.25$		$q = 0.50$
Setting	α	te	all	te	all	te	all	te / all
POPE-1500 LLaVA	0.10	14.9	3.5	14.5	3.5	15.2	3.1	15.5 / 2.8
POPE-1500 LLaVA	0.15	15.5	3.2	14.2	2.0	15.5	2.8	15.2 / 2.8
POPE-adv LLaVA	0.10	10.8	2.5	10.6	3.3	15.3	4.0	16.6 / 3.7
POPE-adv LLaVA	0.15	14.1	0.8	12.6	1.5	12.5	1.3	12.9 / 1.5
POPE-adv Qwen2-VL	0.10	16.0	1.8	15.4	1.8	17.0	3.0	17.6 / 3.6
POPE-adv Qwen2-VL	0.15	10.5	0.0	9.8	0.0	9.9	0.0	9.8 / 0.0
POPE-adv LLaVA+VCD	0.10	13.0	1.9	12.2	2.3	16.3	3.1	**22.3** / **5.1**
POPE-adv LLaVA+VCD	0.15	12.6	2.0	12.8	2.0	14.1	1.5	12.5 / 2.5
AMBER(d) LLaVA	0.10	7.8	1.5	7.2	0.9	8.0	0.9	7.2 / 0.7
AMBER(d) LLaVA	0.15	3.2	0.0	3.2	0.0	3.1	0.0	3.4 / 0.0

A.4 A fully valid alternative to fixing q: three gates at $\delta/3$

§4.6 concedes that fixing $q = 0.10$ a priori is a design choice informed by the same cells that produce the headline. The objection can be removed outright at a price we can quote. Certify all three of $q \in \{0.10, 0.25, 0.50\}$ at $\delta/3$ and select on the calibration fold; the union bound makes this valid without qualification, at the cost of a larger $k_{\min}$ ($22 \to 33$ at $\alpha = 0.10$, $15 \to 21$ at $\alpha = 0.15$).

The $\delta/3$ tax on a *fixed* q is large — 0.8 to 11.8 points of certified coverage for the same $q = 0.10$ — but the union-bound version earns it back by being allowed to select: averaged over the ten cells, the fully valid select-at-$\delta/3$ procedure costs +0.50 points relative to the reported $q = 0.10$-at-full-δ configuration, gaining 1.4 to 7.5 points on five cells and losing 1.4 to 4.2 on the other five. Validity holds throughout (`exc(all)` ≤ 0.012 in every row), with abort rates rising to 12–15% on three cells, which is the visible face of the larger $k_{\min}$.

We report this as an appendix arm rather than adopting it, for one reason worth stating: selecting the q with the largest certified *calibration* coverage systematically prefers the loosest gate — it picks $q = 0.50$ on 61–87% of splits for the two POPE cells — and has no way to know that a loose gate transfers badly across settings (Table 5). A fixed strict gate is preferable to a validly-selected one here because the selection criterion available on the calibration fold is not the criterion that matters, and that is a better argument for fixing q than an unsupported appeal to "fixed a priori".

A.5 Reproducibility

All extracted per-item signals (model probabilities, self-consistency counts, detector scores, labels) are released as JSON, so every table and figure in this paper can be regenerated on CPU without access to a GPU. The GPU scripts that produced those signals are released as well, together with the failed variants discussed in §4.5 and §4.4, which we retain because the failures are part of the argument.

The fixed-sequence procedure lives in one module that every analysis script imports, so a change to the protocol cannot silently apply to some tables and not others. The shared implementation was verified against an independent reference implementation on identical splits across all five settings $\times$ two risk levels $\times$ two arms, with zero disagreements in 2,400 comparisons (identical selected thresholds to 10^{-12}, including agreement on which splits abort). Each table in the paper names the script that produces it in the released manifest, and the two synthetic studies of §A.1 ship as separate scripts.